

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf\_7049e1ed-b93\agent-a9ea9d8cac27b44a5.jsonl`
- SHA-256: `e82a735f429bfba78e7d7fcd91ca819f3fee154753bbc6910f5ff0b0bcc4f2db`
- Source modified: `2026-07-02T04:58:44+00:00`
- Imported at: `2026-07-05T16:48:24+00:00`
- project: `wf\_7049e1ed-b93`
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`

Transcript

1. user (2026-07-02T04:49:27.603Z)

You are a senior engineer producing ONE isolated PR. Today is 2026-07-02.

REPO: C:/Users/avidu/Projects/khelsutra-guru/khelsutra (a shared checkout ? follow the safety rules exactly).

AUDIT FINDINGS YOU ARE FIXING: R1-06, R1-09, R1-36, R1-35 (droplet-LIVE doc rot) ? Grep C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md for these IDs and read the full evidence + verifier notes before coding.

TASK BRIEF:

In C:/Users/avidu/Projects/khelsutra-guru/khelsutra (default branch master; SPA repo ? your PR is DOCS-ONLY, no npm needed), one PR: (1) deploy/DEPLOY_ALPHA.md 'Status: LIVE' ? RETIRED/HISTORICAL banner with the retirement date 2026-06-25, pointer to the GCS archive (gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/) and to issues #97/#114; record the durable decision 'single-origin is the default topology for SPA+CF+Access' (so #97 can be closed); (2) NEXT_STEPS.md:5-13 'Where we are' ? drop the live-droplet/browsable claims, keep the SSH-key revocation as an explicit still-actionable item, re-scope the 'Droplet ? production' queue item to 'any future VM' (site/scripts/provision-vm.sh is provider-agnostic); (3) deploy/GPU_VM_SERVING.md prerequisites ? re-source the Gemini key/AUD/tunnel token from their canonical stores (CF dashboard / owner secret store), not the dead droplet; mark droplet-fallback language historical; (4) docs/LOCAL_APPLIANCE_ARCHITECTURE.md:176 droplet reference ? historical note; (5) deploy/droplet/README (if present) ? retirement banner. VERIFY the retirement facts via git log (khelsutra #117 removed the parity monitor). Do not close issues ? the orchestrator does issue hygiene after merge.

BRANCHING SAFETY (mandatory ? shared checkout, sibling sessions exist):

1. In the MAIN repo checkout run: git fetch origin
2. Create an ISOLATED WORKTREE: git worktree add "C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-<shortname>" -b <branch-name> origin/master (branch explicitly FROM origin/master ? never from the current local branch; use origin/main for repos whose default is main)
3. Do ALL work inside the worktree. Before committing: git branch --show-current + git log --oneline -1 must show YOUR branch on the origin-master base.
4. Stage EXPLICIT paths only (git add <file> <file>) ? NEVER git add -A / . / -u
5. Before opening the PR: git log --oneline origin/master..HEAD must list ONLY your commits.

COMMIT/PR CONVENTIONS:

- Commit message: conventional style matching the repo's history; end the message with a blank line then: Co-Authored-By: Claude Fable 5 <noreply@anthropic.com>
- PR body: what/why, a "## Verification" section stating exactly what you ran locally and what CI covers, and end with: ? Generated with [Claude Code](https://claude.com/claude-code)
- Do NOT merge the PR. Do NOT remove the worktree (a fixer may need it). Return the PR number, branch, worktree path.

QUALITY BAR:

- Read the repo's CLAUDE.md first; for the engine also docs/CODE_MAP.md \$0 (file placement) before adding any new file.

- Add/extend tests for every behavior change. Run the targeted tests you touched (python -m pytest tests/<relevant> -q) + ruff check on changed files; if mypy config exists, run mypy on touched modules. Full suite is CI's job ? do not run it.
- Default-safe: new knobs default OFF / behavior-preserving unless the task explicitly says otherwise.
- The full audit evidence is in C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md ? Grep it for your finding IDs and READ the verifier notes; they contain corrections (exact file:line, scoping traps) you MUST honor.

If the brief's premise turns out false against current code (something already fixed, file moved), adapt honestly and say so in notes ? never force a pointless diff. If you genuinely cannot complete (env broken, auth missing), open NO PR and explain in notes with pr_number=0.

2. user (2026-07-02T04:49:36.437Z)

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97:### R1-06 [P1/S] doc-rot ? khelsutra (avidullu/khelsutra)
124:### R1-09 [P1/S] doc-rot ? khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/
358:### R1-35 [P2/S] hygiene ? khelsutra (avidullu/khelsutra)
367:### R1-36 [P2/S] doc-rot ? khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/
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3. user (2026-07-02T04:49:39.195Z)

master...origin/master

4. user (2026-07-02T04:49:46.737Z)

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85      - verifier notes: Mechanism correction to the evidence: job_worker's five call-time
imports do NOT read db_instance/global_config ? job_worker already takes config/db as explicit
params (job_worker.py:89,205,230,247; mypy.ini:14 notes it was lifted from quarantine) and
resolves FUNCTION seams (ProcessRequest, _execute_compile, _execute_processing, JobWaiting,
_arm_wake_timer, _get_executor, _MAX_DRAIN_WAKE_SEC) through the backend.main module object per
the documented test-monkeypatch contract (job_worker.py:8-16, main.py:286-291). Only
uploads.py:49 reads a main global (global_config). So the app.state migration for db/config is
mostly about uploads + main.py itself; the harder remaining work is redesigning the monkeypatch
seam contract and its tests ? effort is top-end M. Sharper failure scenario: any alternative
entrypoint (new console_script, uvicorn factory) that builds the app without running main()
leaves backend.main.global_config=None and uploads silently falls back to default upload
dir/size limits (uploads.py:49-52), ignoring operator security config ? same class as the
already-documented python -m backend.main incident (main.py:2188-2195) that hung every queued
job. Layering inversion is slightly worse than claimed: the validator's lazy import
(models.py:249) transitively executes segmenters/__init__.py:1-7, importing ALL concrete
segmenters, so constructing AppConfig anywhere drags the full pipeline in; and
SegmenterRegistry.register (base.py:66-70) re-enters AppConfig construction during registration
with errors swallowed by base.py:19's bare except.
86      - verifier reasoning: Verified in current tree (HEAD e410136, pulled clean). Cycle 1:
backend/main.py:292-302 imports backend.api.job_worker at module load (re-exports);
job_worker.py:75,98,214,234,251 do call-time 'import backend.main as _main' (grep-confirmed all
5); uploads.py:47-49 call-time imports and reads _main.global_config; globals defined
main.py:279-280, set at main.py:101-103 and 1981-1982; mypy.ini:15-16 quarantines backend.main,
:22-23 backend.worker.__main__. Cycle 2: config/models.py:244-256 validator imports
segmenter_registry from pipeline/segmenters/base; base.py:14-18 imports AppConfig back in
_normalize_config; importing base initializes segmenters/__init__.py:1-7 which imports every
concrete segmenter (motion.py:243 etc. register as import side effects), so AppConfig
construction depends on pipeline import health ? and register() (base.py:66-70) re-enters
AppConfig via segmenter_cls(None, {}) with failures swallowed by base.py:19 bare except. Eval
cycles lazy+documented (golden_fixtures.py:1008-1013). Not already fixed: git log for the four
files shows only feature commits (#405-#413). Severity honest: the failure class already bit ?
main.py:2188-2195 documents the python -m backend.main dual-module incident
(backend.main.db_instance=None, drain crashed silently, jobs hung 'queued' forever), patched
only for that entry path (main.py:2196-2198); uploads.py:49-52 getattr-defaults mean any
entrypoint that skips main() silently runs uploads with default dir/limits. Tracked_in accurate:
audit doc line 221 is verbatim the P5 shim note. Not owner-gated: only owner question is D1
```

phase sequencing, none of the listed gates. Fix sane: create_app exists (main.py:2184), handlers already use app.state (mypy.ini:12-13), backend/config/startup.py exists as the startup-check home.

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87
88     ### R1-05 [P1/M] tech-debt ? badminton-highlight-indexer
89     **Path-jail rejects gs://,s3://,gdrive-api:// before scheme resolution ? bucket ingest
impossible on the exposed alpha box (issue #327 still open)**
90     - anchor: `backend/main.py:1376` · tracked in: https://github.com/Khelsutra/badminton-highlight-indexer/issues/327 · finder: engine:security-config
91     - evidence: main.py:1376 calls `validate_input_path(req.video_path,
allowed_root=allowed_root)` BEFORE `storage.resolve_input_ref` at main.py:1379;
backend/utils/paths.py:45-55 `resolve_under_root` realpaths the raw URI so `gs://bucket/x.mp4`
escapes the jail -> 400. Startup check startup.py:106-119 makes the jail mandatory when exposed
(EXPOSED_NO_VIDEO_ROOT), so an exposed engine can never ingest from a bucket. gh: issue #327
OPEN (verified 2026-07-02), proposing scheme-aware validation.
92     - why it matters: Blocks the alpha CUJ (large-video ingest via gs:// instead of 500 MB
browser uploads) and is a stale-open thread since before 06-26; the fix must be done carefully
to avoid reopening the arbitrary-local-file hole the jail exists for.
93     - proposed fix: Implement #327 option (b): jail only after resolve_input_ref, applying
resolve_under_root exclusively to backend=='local' refs; keep traversal/symlink/absolute
rejection for local paths and add both allow (bucket URI accepted) and reject (local traversal
still 400) tests, updating
tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri.
94     - verifier notes: Minor corrections: (1) startup check path is
backend/config/startup.py:106-119, not startup.py (lines exact). (2) The "500 MB browser
uploads" figure in why_it_matters comes from issue #327's body; the repo's actual browser-upload
cap is security.max_upload_mb default 4096 MB (backend/api/uploads.py:51,
backend/config/models.py:339) ? 500 MB is the Gemini provider per-clip cap
(backend/providers/gemini.py:23). Immaterial to the finding (multi-hour matches exceed 4 GB
anyway; docs/VIDEO_LOCALITY_MODEL.md:26 "The 10 GB problem is real"). (3) Fix scope additions:
there is a second resolve call-site (CLI --video-path, see tests/test_api_endpoints.py:922) ?
verify whether it needs the same scheme-aware treatment; ensure unknown schemes still
ValueError?400 via resolve_input_ref as the sole scheme gate. (4) Caution, not a gate: the
change relaxes a deliberately pinned security contract on the exposed box; #327 records the
owner's explicit want, but the option (a) vs (b) choice and the PR should get owner sign-off
before landing since it alters exposed-serving posture.
95     - verifier reasoning: Personally reproduced from current code. (1) Ordering:
backend/main.py:1376 calls validate_input_path(req.video_path, allowed_root=allowed_root) BEFORE
storage.resolve_input_ref at main.py:1379-1381; ValueError maps to HTTP 400 at
main.py:1382-1383. (2) Mechanism: backend/utils/paths.py:58-68 routes any request with
allowed_root set through resolve_under_root, which os.path.realpath()s the RAW string
(paths.py:47) ? 'gs://bucket/x.mp4' realpaths to a cwd-relative path, commonpath != root
(paths.py:50-54) ? ValueError ? 400. The code itself documents this at main.py:1387-1389: "a
configured allowed_video_root (hosted mode) rejects a non-local URI as out-of-root". (3) Pinned
by test: tests/test_api_endpoints.py:903-912 asserts gs://, s3://, gdrive:// all ? 400 when
allowed_video_root is set ("Pins the security contract"). (4) Exposure mandate:
backend/config/startup.py:106-119 emits LEVEL_ERROR EXPOSED_NO_VIDEO_ROOT when edge_auth_enabled
and allowed_video_root unset ? so an exposed engine must have the jail, making bucket ingest
impossible there. (5) Not fixed: git log -15 -- backend/main.py (head e410136, #413) shows no
commit addressing this; behavior present in current tree. (6) gh issue view 327: state OPEN,
created 2026-06-22, title/body match the finding exactly, including the owner's explicit want
("ingest large videos via gs:// on the hosted box") and options (a)/(b). Concrete failure:
exposed alpha box (edge_auth_enabled=true, allowed_video_root=/jail per boot requirement)
receives POST /api/process {"video_path":"gs://bucket/match.mp4"} ? 400 "path ... escapes the
allowed root"; only ingest path left is the capped browser upload (backend/api/uploads.py:51,
security.max_upload_mb) ? blocks the tracked alpha large-video CUJ. Severity P1 honest (owner-
wanted CUJ blocked, open since 2026-06-22, workaround capped/painful per
docs/VIDEO_LOCALITY_MODEL.md:19-26). Proposed fix (option b: jail only backend=='local' after
resolve_input_ref, keep null-byte check universal and traversal/symlink jail for local, add
allow+reject tests, update the pinned test) matches the issue's own proposal and is correctly
scoped; effort M is right.
96
97     ### R1-06 [P1/S] doc-rot ? khelsutra (avidullu/khelsutra)
98     **Docs claim the retired droplet alpha is LIVE (droplet deleted 2026-06-25)**
99     - anchor: `NEXT_STEPS.md:12` · tracked in: partially in issue #97 (item 2: update
```

DEPLOY_ALPHA.md); the NEXT_STEPS.md droplet claim is untracked · finder: khelsutra:issues-triage

100 - evidence: NEXT_STEPS.md:12-13 ('Where we are (updated 2026-07-01)') still says the site is 'live + browsable on the D0 droplet -> http://168.144.126.176:8787'; deploy/DEPLOY_ALPHA.md:3 says 'Status: LIVE (deploy live-verified on the droplet 2026-06-22)' and :29 'This runbook ships the split topology'; docs/LOCAL_APPLIANCE_ARCHITECTURE.md:176 'The alpha stand-up is now LIVE (2026-06-22)'. Project memory (MEMORY.md:63, session-handoff.md:19): droplet claude-do-rally-test/168.144.126.176 RETIRED-deleted 2026-06-25; alpha paused pending Cloud Run. nslookup shows api./app.khelsutra.guru still resolve via Cloudflare to a tunnel with no origin (pending CNAME/Access cleanup).

101 - why it matters:

102 - proposed fix: One doc PR: mark DEPLOY_ALPHA.md status RETIRED/PAUSED with a pointer to the Cloud Run plan, fix NEXT_STEPS.md 'Where we are' (droplet gone, site is CF-Workers-only), and correct LOCAL_APPLIANCE_ARCHITECTURE.md:176. Also note the single-origin repoint that superseded the split topology (issue #97). The 2026-07-01 refresh (#139) touched docs but kept the stale claim, so this will keep misleading ramp-up reads.

103 - verifier notes: Minor anchor correction: the 'Where we are (updated 2026-07-01)' header is NEXT_STEPS.md:5; the droplet-live claim spans lines 12-13 (the finding's line 12 anchor is correct for the URL text). Severity P1 stands: NEXT_STEPS.md's 2026-07-01 stamp is NEWER than the handoff's 2026-06-25 correction, so ramp-up readers reconciling the two will trust the wrong doc; worse, DEPLOY_ALPHA.md \$1 (line 35) directs deploying the engine + secrets onto 168.144.126.176 ? an IP the owner no longer controls and D0 can recycle. Fix is sane, doc-only, and not owner-gated (it records already-made owner decisions: retirement 2026-06-25 + single-origin repoint per issue #97); one caution ? the fix should point to the Cloud Run move as 'paused pending' rather than declare a new serving decision, which would be owner-gated. Did not independently re-verify the finder's nslookup/DNS claim; the finding stands without it.

104 - verifier reasoning: Working tree at HEAD f9eea14: NEXT_STEPS.md:5 'Where we are (updated 2026-07-01)' + :12-13 'live + browsable on the D0 droplet -> http://168.144.126.176:8787 (systemd khelsutra-site)'; deploy/DEPLOY_ALPHA.md:3 'Status: LIVE (deploy live-verified on the droplet 2026-06-22...)' + :29 'This runbook ships the split topology'; docs/LOCAL_APPLIANCE_ARCHITECTURE.md:176 'The alpha stand-up is now LIVE (2026-06-22): the engine booted on the droplet (168.144.126.176)'. Ground truth: memory/khelsutra-site-droplet-parity.md:3,10-14 (droplet DELETED 2026-06-25, data archived, doctl delete done, monitor removed via PR #117), session-handoff.md:19 ('the prior LIVE alpha on the droplet is GONE ? alpha paused pending the Cloud-Run move'), MEMORY.md:63. Not fixed: git log shows the last commit touching all three files is 9223ba9 (#139, the 2026-07-01 refresh) and the stale text persists at HEAD. Tracking verified: gh issue 97 is OPEN, item 2 covers only DEPLOY_ALPHA.md (single-origin update); the NEXT_STEPS.md droplet claim is untracked. Issue #97 also confirms split topology was superseded by the 2026-06-22 single-origin repoint, so DEPLOY_ALPHA.md:29 is doubly stale.

105

106 ### R1-07 [P1/M] critical-fix ? khelsutra (avidullu/khelsutra)

107 **Issue #111 still fully valid: refresh loses the in-flight job; progress static for minutes**

108 - anchor: `web/src/hooks/useIngestJob.ts:117` · tracked in: khelsutra#111 · finder: khelsutra:issues-triage

109 - evidence: useIngestJob.ts:117-128 ? job_id lives only in the poll closure (no localStorage/URL persistence); App.tsx:71-73 seeds state only from ?video= (set on done, App.tsx:153). A refresh mid-job lands the user back on the ingest form while the server job keeps running. Progress is the coarse free-text job.progress (useIngestJob.ts:106); no heartbeat/animation beyond a spinner chip (IngestPanel.tsx:337-358).

110 - why it matters:

111 - proposed fix: SPA half is Claude-doable now: persist the active job_id (?job=<id> + localStorage) and on load resume polling to the existing ?video= handoff; add elapsed-time/heartbeat copy. Engine half (granular 'segmenting chunk 3/7' from backend/pipeline/segmenters/gemini.py via the job-progress hook) is a small cross-repo change. Per the issue this is 'the most likely way an early tester loses a working job'.

112 - verifier notes: Failure scenario, sharpened from issue #111's field report: an ~8-min upload spends ~5 min in Gemini segmentation with near-static "processing" text; the tester refreshes (natural when nothing moves), lands on the ingest form with no trace of the running job, and either re-submits ? a duplicate job, potentially on paid cloud GPU via the compute picker ? or abandons believing it failed; the finished video only appears later in the library with no pointer back. Fix note: the backend is already refresh-safe (jobs persisted in SQLite), so the SPA half (persist ?job=/localStorage, resume polling into the existing ?video= handoff, heartbeat/elapsed-time copy) is standalone and Claude-doable now; only the granular-progress half touches the sibling badminton-highlight-indexer repo.

113 - verifier reasoning: useIngestJob.ts:117-128 ? job_id is a closure-local (`let jobId` at :117, used only by poll(:128)); grep of web/src confirms no localStorage/URL persistence of any job id (only annotate flag, wizard key, i18n lang). App.tsx:71-73 seeds videoId solely from ?video=, written only on job-done via loadVideo (App.tsx:150-154 ? useIngestJob.ts:96-98), so a mid-job refresh renders the ingest form (App.tsx:338-343) while the server job keeps running. Progress is coarse free-text job.progress (useIngestJob.ts:106) shown as a spinner chip (IngestPanel.tsx:337-348), no heartbeat/elapsed time. git log -15 on both files (latest #140, #138, #128?) shows no fix; issue khelsutra#111 is OPEN and documents the identical gaps from a live 2026-06-22 alpha run. Engine path backend/pipeline/segmenters/gemini.py exists in the sibling badminton-highlight-indexer repo as the finder scoped it. Not owner-gated: pure client resiliency, no go-live/corpus/licensing/spend/legal decision.

114

115 ### R1-08 [P1/S] tech-debt ? khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/
116 **Engine's vendored SPA bundle is 8 feature-PRs stale ? shipped appliance UI lacks the entire storage/annotation wave**

117 - anchor: `badminton-highlight-indexer/backend/ui/ui\_version.json:3` · tracked in: ? · finder: khelsutra:code

118 - evidence: backend/ui/ui_version.json pins spa_commit ed33a515 built_at 2026-06-22T17:59:59Z, but khelsutra web/src has 8 newer PRs all merged 2026-07-01 (git log: f9eea14 #140 fps-in-library, fb2f561 #138 FFmpeg banner, 8667dbf #137 medium picker, cc06c32 #128 AnnotatePanel, 9e9df72 #127 persistent name, 926b53a #126 vendored annotator, 39c1349 #124 library, d0285cd #122 e2e harness). grep for 'video-library'/'library.title' in backend/ui/assets/index-ukuS9wxa.js = 0 hits. Meanwhile the engine ALREADY serves the matching endpoints: /api/assets (backend/main.py:727), /api/source (main.py:623), /api/annotations (main.py:663), and PRs #411-413 (e410136, 81a78bf) enriched /api/videos specifically for the new SPA library ? so a wheel install serves an API whose UI can't reach its own features. The API_VERSION skew guard doesn't fire (still 1, main.py:408); nothing detects content staleness.

119 - why it matters:

120 - proposed fix: Re-run badminton-highlight-indexer/scripts/bundle_ui.sh against khelsutra@f9eea14 and commit the refreshed backend/ui/ (the script's own convention: 'Refresh the snapshot by re-running this on an SPA release; commit the diff'). Then add a cheap staleness signal: a CI/doctor check comparing ui_version.json spa_commit against khelsutra origin/master (warn when >N commits behind), so the bundle-sync convention can't silently rot again.

121 - verifier notes: Minor corrections only: API_VERSION is at backend/main.py:407, not 408; and web/ actually has 10 commits since the pin (the 8 cited feature/test PRs plus docs #139/#123). Sharper failure scenario: a user who pip-installs the wheel and runs `indexer --app` gets an appliance UI with no video library, no annotate panel, no storage-medium picker, and no FFmpeg-missing banner, while the very same wheel serves /api/assets, /api/source, /api/annotations, and the just-enriched /api/videos (fps/resolution/match_name from #412/#413) ? the engine's two newest commits are dead payload for every wheel user. P1 stands because both repos' latest work (engine #411-413, SPA P5-P8) targets features unreachable in the only shipped artifact. Fix scope is right; note the CI staleness check needs network access to the khelsutra remote, so a doctor-time warning may be the cheaper variant.

122 - verifier reasoning: backend/ui/ui_version.json:3 pins spa_commit ed33a515 (built_at 2026-06-22T17:59:59Z, line 6); khelsutra `git log ed33a515..HEAD -- web/` shows all 8 cited PRs (#140 f9eea14, #138 fb2f561, #137 8667dbf, #128 cc06c32, #127 9e9df72, #126 926b53a, #124 39c1349, #122 d0285cd) merged 2026-07-01. Vendored bundle backend/ui/assets/index-ukuS9wxa.js has 0 occurrences of 'video-library' (the 2 'Your videos' hits are privacy-policy prose), while current web/src/components/VideoLibrary.tsx:44,48 uses t("library.title") and data-testid="video-library". Engine endpoints exist as claimed: /api/source backend/main.py:623, /api/annotations :663, /api/assets :727, and engine HEAD e410136/#413 + 81a78bf/#412 enriched /api/videos for the new library. API_VERSION = 1 at backend/main.py:407 with a comment that additive endpoints need no bump, so the skew guard cannot fire; no spa_commit staleness check exists in backend/. Not fixed: last commit touching backend/ui/ is 8bcf6ce (2026-06-22). scripts/bundle_ui.sh header confirms the refresh-and-commit convention the fix proposes. Not owner-gated: documented routine snapshot refresh; AnnotatePanel is flag-gated in the SPA (#128) so bundling is safe.

123

124 ### R1-09 [P1/S] doc-rot ? khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/
125 **DEPLOY_ALPHA.md claims 'Status: LIVE' on a droplet retired 2026-06-25; GPU_VM_SERVING.md sources its secrets from that droplet**

126 - anchor: `khelsutra/deploy/DEPLOY\_ALPHA.md:3` · tracked in: khelsutra #97, #114 (topology); droplet retirement itself untracked in-repo · finder: khelsutra:code

127 - evidence: deploy/DEPLOY_ALPHA.md:3 `\*\*Status:\*\* `LIVE` (deploy live-verified on the droplet 2026-06-22...); :35 names 'the CPU droplet 168.144.126.176' as the engine box. The

droplet was retired 2026-06-25 (khelsutra commit 5ffb750 #117: 'retire the live-parity monitor ? droplet mirror decommissioned'); grep -i 'retired|decommission' over the repo's docs = 0 hits, so no doc records it. deploy/GPU_VM_SERVING.md §0 tells the operator to copy the CF-Access AUD from 'the droplet's /srv/khelsutra/config.json' and the tunnel token from 'systemctl cat cloudflared' on the droplet, and calls the droplet a 'hot fallback' ? all impossible now. Secondary rot: DEPLOY_ALPHA.md:143 claims the SPA 'shows a friendly engine offline banner (/api/health ping)' but web/src contains zero references to /api/health (grep -i health = no matches); and open issue #114 (Access sign-in broken) + #97 (split-origin breaks CF Access) contradict the runbook's shipped topology.

128 - why it matters:

129 - proposed fix: One doc PR: flip DEPLOY_ALPHA.md status to RETIRED/HISTORICAL with a pointer to the retirement date and to issues #97/#114; rewrite GPU_VM_SERVING.md prerequisites to source the Gemini key/AUD/tunnel token from their canonical stores (Drive / CF dashboard) instead of the dead droplet; delete or soften the unimplemented /api/health-banner claim. The related api.khelsutra.guru CNAME + CF-Access application cleanup is Cloudflare-dashboard work (owner-gated) ? note it in the doc as pending.

130 - verifier notes: Two refinements. (a) "all impossible now" is slightly overstated: per 5ffb750's message the droplet's data was archived to gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/, so /srv/khelsutra/config.json (and its cf_access_aud) is likely recoverable from that archive, and the tunnel token is recoverable from the Cloudflare dashboard since the tunnel is token-managed (GPU_VM_SERVING.md:24-26). The doc rewrite should point at the gs:// archive + CF dashboard as the canonical sources ? strictly better than the finder's "Drive / CF dashboard" suggestion for the AUD. (b) DEPLOY_ALPHA.md:143 is partial rot, not total: the SPA does degrade gracefully when the engine is unreachable (SetupWizard offline branch, web/src/components/SetupWizard.tsx:84-142), but via a capabilities fetch, and there is no persistent "engine offline" banner and zero /api/health references ? so soften rather than delete the claim. Also worth citing commit 5ffb750 / PR #117 and the "Cloudflare is now the single canonical host" line in the doc PR as the retirement record, since no .md currently captures it.

131 - verifier reasoning: Personally reproduced every element of the claim in the current working tree (khelsutra @ f9eea14). (1) deploy/DEPLOY_ALPHA.md:3 reads "***Status:** `LIVE` (deploy live-verified on the droplet 2026-06-22...)" and :35 names "the CPU droplet `168.144.126.176`" as the engine box; :143 claims the SPA "shows a friendly 'engine offline' banner (`/api/health` ping)". (2) Commit 5ffb750 (2026-06-25, PR #117) states verbatim: "That droplet has been decommissioned (its data archived to gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/)... Cloudflare is now the single canonical host" ? it only removed .github/workflows/parity.yml + site/scripts/parity-check.sh, touching neither deploy doc. (3) grep -i 'retired|decommission' over all *.md in the repo = 0 matches, so no doc records the retirement. (4) deploy/GPU_VM_SERVING.md:24 still says "the droplet stays as a hot fallback"; :37-40 instruct sourcing the CF-Access AUD from "the droplet's /srv/khelsutra/config.json" and the tunnel token from "systemctl cat cloudflared" on the droplet ? and that file was last edited 2026-06-29 (ca8bfcc #119), FOUR DAYS AFTER the decommission, without fixing this, so NOT already-fixed. (5) grep over web/src: zero references to /api/health anywhere (only FfmpegBanner.tsx exists as a banner; SetupWizard.tsx:84-142 has an offline message driven by a capabilities fetch, not a /api/health ping), confirming the :143 rot. (6) gh: issues #114 (allowlisted email couldn't sign in, real demo 2026-06-23) and #97 (split-origin breaks CF Access ? single-origin needed) are both OPEN, contradicting "Status: LIVE ... only a rotated Gemini key + a browser OTP login remain". Concrete failure: this is a docs-as-memory, multi-session project ? the next serving session following GPU_VM_SERVING.md §0 is blocked at prerequisites trying to SSH a nonexistent box for secrets, and DEPLOY_ALPHA.md's LIVE status + "hot fallback" cutover-safety claim would drive wrong go-live/rollback decisions for the actual alpha. P1 is honest for the primary serving runbook; effort S (one doc PR) is right. The doc PR itself is not owner-gated; only the Cloudflare-dashboard CNAME/Access cleanup is, and the proposed fix already scopes that as a note.

132

133 ### R1-10 [P1/S] hygiene ? sports-data-collector

134 **Stray nested clone sports-data-collector/sports-data-collector/ pollutes tooling; safe to delete**

135 - anchor: `sports-data-collector/sports-data-collector/:` · tracked in: ? · finder: sdc:audit

136 - evidence: Untracked dir is a FULL second clone of the same repo: remote =

<https://github.com/avidullu/sports-data-collector.git> (pre-org-move URL; parent origin is

<https://github.com/Khelsutra/sports-data-collector.git>). HEAD = 031e410 'feat(annotation):

richer Roora schema (#29)' (~Jun 19; working-tree files mtime Jun 19 21:38, .git mtime Jun 29

14:41 from a later fetch that moved its origin/master to a152cf5). Verified zero unique work:

`git merge-base --is-ancestor 031e410 HEAD` in the parent succeeds, `git status --short` clean, `git stash list` empty, only branch is master. Size ~1.7 MB (811 KB .git). It contains its own tests/, src/, .github/, and data/sports_metadata.db (tracked at that commit, unmodified). Tooling pollution measured: `ruff check .` from repo root reports 39 errors vs 24 when scoped to src/tests/cleanup.py ? the extra 15 come from the nested clone; bare `pytest` rootdir collection and setuptools package discovery would also pick it up.

137 - why it matters:

138 - proposed fix: Delete the entire nested directory (rm -rf sports-data-collector/sports-data-collector/). Nothing in it is unique: HEAD is an ancestor of current master, tree is clean, no stashes/branches. Root cause is almost certainly a `git clone` run with cwd already inside the repo (~Jun 19); worth adding `sports-data-collector/` to .git/info/exclude only if deletion is deferred. READ-ONLY audit ? not deleted; owner or a follow-up session should remove it.

139 - verifier notes: Sharper failure scenario than the finder's: beyond the +15 phantom ruff errors, both tests/ dirs share module basenames with no __init__.py, so bare `pytest` from repo root fails at collection with import-file-mismatch errors ? i.e., the owner's pre-LGTM gate ("full test suite green", "ruff check . clean") is actively corrupted for any local session until the dir is deleted; agents using root-wide Grep/Glob also risk editing the stale nested copy. Two cautions for the fixer: (a) parent working tree currently has an unrelated modified .gitignore (adds `artifacts/` ? likely sibling/owner WIP per global instructions; do not clobber or commit it while deleting the nested dir); (b) nested .git mtime is Jul 2 08:18 (today) ? something (likely the finder's fetch or background maintenance) touched it, but no new local work exists (origin/master..master empty), so deletion remains safe.

140 - verifier reasoning: Every material claim reproduced against the current tree. (1) Nested clone exists: C:/Users/avidu/Projects/khelsutra-guru/sports-data-collector/sports-data-collector/ is a full checkout (.git, src/, tests/, .github/, data/, cleanup.py; dir mtime Jun 19 21:38) and parent `git status --short` shows it untracked (`?? sports-data-collector/`). (2) Remotes match the claim: nested `git remote -v` = https://github.com/avidullu/sports-data-collector.git (pre-org-move), parent origin = https://github.com/Khelsutra/sports-data-collector.git. (3) Zero unique work, personally re-verified: nested HEAD = 031e410 "feat(annotation): richer Roor schema (#29)"; `git merge-base --is-ancestor 031e410 HEAD` in the parent (HEAD adcd206) succeeds; nested `git status --short` clean, `git stash list` empty, only branch master, and `git log origin/master..master` empty (no unpushed commits). Size matches: du = 1.7M total, 811K .git. (4) Tooling pollution reproduced exactly: `ruff check . --no-cache` from repo root = "Found 39 errors" vs `ruff check src tests cleanup.py --no-cache` = "Found 24 errors" ? the 15-error delta is the nested clone. No guard exists: pyproject.toml (lines 1-41, entire file) has no [tool.ruff] or [tool.pytest.ini_options] section, and no pytest.ini/setup.cfg/tox.ini/ruff.toml anywhere; nested dir absent from .gitignore and .git/info/exclude. (5) The pytest claim is structurally worse than stated: tests/ and sports-data-collector/tests/ contain identically-named modules (test_batch_preflight.py, test_cli.py, test_config.py, test_cvat_parser.py, ...) and NEITHER has __init__.py, so a bare `pytest` from root would hit "import file mismatch" collection errors ? breaking the owner's mandated full-suite pre-LGTM gate locally. Not owner-gated: pure local hygiene, no serving/training/licensing/spend/ToS surface. Fix (rm -rf the nested dir) is safe and correctly scoped given the ancestor/clean/no-stash/no-unpushed proof.

141

142 ### R1-11 [P2/M] ci-infra ? badminton-highlight-indexer

143 **ci.yml/nightly.yml lane drift predicted by #384 has now materialized (link-check missing from nightly, timeout divergence, duplicated coverage floor)**

144 - anchor: `.github/workflows/nightly.yml:12` · tracked in:

https://github.com/Khelsutra/badminton-highlight-indexer/issues/384 · finder: engine:ci-tests

145 - evidence: ci.yml:53-63 added a link-check job (commit af1ccbe, PR #402, 2026-06-30) that nightly.yml does not mirror; nightly.yml:12-16 still claims 'every OFFLINE guard job from ci.yml is mirrored here' and its notify-on-failure needs list (nightly.yml:234) has no link-check; timeouts have also diverged (license-scan: ci.yml:187=15min vs nightly.yml:171=10min; build-artifact: ci.yml:214=30min vs nightly.yml:197=20min); coverage floor 70 is a duplicated literal (ci.yml:145, nightly.yml:139) that nightly.yml:22-23 itself flags as not single-sourced

146 - why it matters: Issue #384 predicted exactly this: copied lanes silently drift. The drift is no longer hypothetical ? the nightly's parity claim is false, and the next coverage-floor ratchet in ci.yml would leave nightly enforcing a stale floor with nobody noticing (the nightly's whole purpose is catching rot during quiet periods).

147 - proposed fix: Do the #384 refactor now: extract the shared offline guard lanes into .github/workflows/_offline-guards.yml exposed via workflow_call with the coverage floor as an input, and have both ci.yml and nightly.yml call it (add link-check to the shared set or document why it is per-PR-only); fix the stale parity comment in nightly.yml:12-16 either way.

148 - verifier notes: Two corrections. (a) The timeout "divergence" is congenital, not

materialized drift: ``git merge-base --is-ancestor`` shows 3a34c42 (#382, ci.yml timeouts 15/30) landed BEFORE the nightly was created (9972057, #381), and ``git show 9972057:nightly.yml`` already carried 10/20 ? the values never matched, so only the link-check omission + stale parity comment are new drift from af1ccbe. (b) Severity downgraded P1?P2: the sharpest real failure scenario today is weak ? tools/check_md_links validates internal repo links, which only change via commits, and every push/PR runs ci.yml's link-check on identical content, so the nightly missing that lane forfeits ~no detection ("rot during quiet periods" cannot affect a content-only check). The coverage-floor stale-ratchet failure is future-conditional (requires a ratchet PR that forgets nightly.yml:139), and the nightly's tighter license-scan timeout (10 vs 15) would at worst cause a visible spurious failure (notify-on-failure opens a tracking issue ? noisy, not silent). What IS immediate is a false parity comment at nightly.yml:12-16. Proposed fix is sane and matches open issue #384; minimum viable slice if the M-sized refactor is deferred: add link-check to nightly + fix the comment + align timeouts (S).

149 - verifier reasoning: Verified in current tree (master, up to date with origin). (1) ci.yml:53-63 defines the link-check job (runs ``python -m tools.check_md_links``, offline), added by commit af1ccbe (PR #402, 2026-06-30) per ``git log -- .github/workflows/ci.yml``; ``git show --stat af1ccbe`` confirms it touched ci.yml but not nightly.yml. (2) nightly.yml:12-16 still asserts "every OFFLINE guard job from ci.yml is mirrored here" listing only 8 lanes (no link-check) ? the parity claim is now false; nightly.yml:234 needs list ``[lint, leak-scan, typecheck, import-smoke, full-suite, golden-fixtures, license-scan, build-artifact]`` also lacks link-check. (3) Coverage floor is a duplicated literal: ci.yml:145 and nightly.yml:139 both ``--cov-fail-under=70``; nightly.yml:22-23 self-flags "still copied literals, not single-sourced". (4) Timeouts differ as cited: license-scan ci.yml:187=15 vs nightly.yml:171=10; build-artifact ci.yml:214=30 vs nightly.yml:197=20. (5) Not fixed: nightly.yml last commit is ffb1f53 (#391); no `_offline-guards.yml` or any `workflow_call` reusable exists (.github/workflows/ contains only ci.yml and nightly.yml). (6) ``gh issue view 384`` = OPEN, proposing exactly the `workflow_call` extraction; its own triage header says "LOW ? drift". Not owner-gated: pure CI plumbing, floor stays at 70 (respects the ratchet-only note at ci.yml:141-143), no serving/corpus/licensing/spend decisions.

150

151 ### R1-12 [P2/S] security ? badminton-highlight-indexer

152 **Default CORS allow_origins=['*'] lets any website script the localhost appliance (paid Gemini runs + local data read)**

153 - anchor: ``backend/main.py:116`` · tracked in: ? · finder: engine:security-config

154 - evidence: backend/main.py:114-116 ``_cors_origins_from_env`` returns ``origins or ["*"]``; main.py:123-129 adds `CORSMiddleware` with `allow_methods=['*']`, `allow_headers=['*']`. The `TrustedHost` guard (main.py:132-156) only blocks DNS-rebinding ? a direct fetch to `http://127.0.0.1:xport>` from a hostile page sends `Host: 127.0.0.1` (allowed) and the wildcard CORS answers its preflight, so evil.com JS can `POST /api/process` (no auth by default: backend/api/auth.py:126-127 returns `None` when `edge_auth` off) and trigger a paid Gemini run (main.py:1817 reads `GEMINI_API_KEY`; issue #340 documents re-billing) and read `/api/videos` responses. The UI is served same-origin (main.py:240,255 `StaticFiles` mounts), so wildcard CORS is not needed for the primary UX. Mitigated in current Chrome by Local Network Access prompts, NOT in Firefox.

155 - why it matters: Cost-leak plus local-data exposure reachable in normal use: a user browsing any hostile page while the appliance runs can have their Gemini key billed and their video index read. It silently defeats the P0c DNS-rebinding hardening the repo already invested in.

156 - proposed fix: Change the default to same-origin/loopback (e.g. ``[]`` plus skipping the middleware, or an explicit localhost dev-origin list) and keep `RALLY_CORS_ALLOW_ORIGINS` as the widening knob; add a startup exposure/appliance warning when the wildcard is active. Add a regression test that an evil-origin preflight is rejected by default.

157 - verifier notes: Downgrade P1?P2: the failure scenario is real but the finding itself honestly notes Chromium Private Network Access mitigates it in Chrome/Edge (dominant browser share); residual exposure is mainly Firefox against a local single-user appliance. Still a legitimate secure-default regression worth fixing ? it silently undercuts the P0c hardening and is inconsistent with the repo's own loopback-by-default posture for bind host and allowed hosts. Fix and effort (S) as proposed are accurate.

158 - verifier reasoning: All cited code exists and behaves as claimed in current HEAD (e410136). backend/main.py:109-116 ``_cors_origins_from_env`` returns ``origins or ["*"]``; main.py:123-129 adds `CORSMiddleware` with `allow_origins` from that fn, `allow_methods=['*']`, `allow_headers=['*']`, `allow_credentials=False`. main.py:152-156 `TrustedHostMiddleware` defaults to loopback hosts and matches on `host.split(':')[0]`, so a direct fetch to `http://127.0.0.1:<port>` sends `Host:127.0.0.1` and passes ? it only stops DNS-rebinding via attacker domains, not direct-IP cross-origin fetches. backend/api/auth.py:125-127 `resolve_tenant` returns `None` (no auth) when

edge_auth off (default), and /api/process (main.py:1355-1370) has no other auth. Mechanism is correct: wildcard CORS is load-bearing to (a) read /api/videos responses cross-origin and (b) pass the JSON-preflight for POST /api/process (main.py:1817 ? paid Gemini). /api/process 404 fast-fail (main.py:1390) needs a valid local path, but that is harvestable from the readable /api/videos index, making the cost-leak chain concrete. git log -15 -- backend/main.py shows no fix; wildcard default is live. The finding also correctly flags the internal inconsistency: bind host (main.py:174) and allowed hosts (main.py:146) both default to loopback ('widen consciously'), yet CORS alone defaults open ? an honest oversight. Proposed fix is sane and correctly scoped; not owner-gated (local-appliance secure default, no cloud-strategy/serving-go-live decision).

159

5. user (2026-07-02T04:49:47.567Z)

350 **Issue #112 still valid: Share to WhatsApp sends text only, no Web Share API attempt**
351 - anchor: `web/src/components/CompileResult.tsx:79` · tracked in: khelsutra#112 ·
finder: khelsutra:issues-triage
352 - evidence: CompileResult.tsx:79 still builds the wa.me text-only link ('wa.me can't attach the local file' comment at :76-77); grep for navigator.share/canShare over web/src returns no matches ? the proposed file-share path was never started. Labeled 'bug' on the issue; reel URLs are CF-Access-gated so a link can't substitute.
353 - why it matters:
354 - proposed fix: Implement the issue's plan: fetch the reel blob -> new File -> navigator.canShare({files}) ? navigator.share(...) : graceful fallback (download + 'attach it in WhatsApp' hint, keep wa.me). Contained to CompileResult + catalog strings + tests; main win is mobile one-tap video share.
355 - verifier notes: Failure scenario, concrete: alpha user compiles a reel on mobile, taps "Share to WhatsApp" (CompileResult.tsx:78-86) ? WhatsApp opens with pre-filled text and no video; user perceives Share as broken (reported from live alpha 2026-06-22 per issue body), and any URL they paste instead 403s for recipients behind CF Access. P2 is honest ? degraded share UX with a manual workaround (download + attach), not a blocked core flow. Effort M is right: blob fetch ? new File ? canShare gate ? share, plus fallback copy, catalog strings (result.shareText retained), and component tests; note desktop Edge (where the bug was hit) will likely still land on the fallback since Web-Share file support there is limited ? the win is mobile one-tap video share, as both the finding and issue #112 state.
356 - verifier reasoning: web/src/components/CompileResult.tsx:76-77 carries the comment "wa.me can't attach the local file, so the user adds the just-downloaded reel", and :78-79 builds the text-only link `https://wa.me/?text=\${encodeURIComponent(t("result.shareText"))}` ? exactly as claimed. Grep for navigator.share / navigator.canShare / webShare (case-insensitive) across web/ returns zero matches, so no Web Share API attempt exists anywhere in the SPA. git log for the file shows the newest commit touching it is 0fae364 "feat(web+site): A2 ? localized WhatsApp share for the reel (#82)" ? the commit that ADDED the wa.me link; nothing since fixed it (working tree clean on master...origin/master). gh issue view 112 (avidullu/khelsutra): state OPEN, labeled "bug", body matches the finding including the CF-Access rationale. That rationale is corroborated in-repo: deploy/GPU_VM_SERVING.md:37-38 and :87 show serving sits behind Cloudflare Access (cf_access_aud / cf_team_domain, "auth = the Cf-Access JWT verify"), so a shared reel URL is unreachable for recipients outside the tenant ? the file-share path is the only real fix. Not owner-gated: contained frontend UX change implementing the owner's own written plan in #112; no go-live, corpus/training, spend, or legal surface.

357

358 ### R1-35 [P2/S] hygiene ? khelsutra (avidullu/khelsutra)
359 **Issue #97 stale-open: describes retired infrastructure; topology decision needs re-scoping to the Cloud Run alpha**
360 - anchor: `deploy/DEPLOY\_ALPHA.md:29` · tracked in: khelsutra#97 · finder: khelsutra:issues-triage
361 - evidence: The issue's 'fix applied' state (engine-served single-origin on the droplet tunnel) no longer exists: droplet retired/deleted 2026-06-25 (memory MEMORY.md:63; session-handoff.md:19 'alpha is paused pending the Cloud-Run move; api.khelsutra.guru CNAME/CF-Access need cleanup'). Its remaining items are stale: (1) app. DNS repoint targets a tunnel that is gone; (3) 'refresh the bundled SPA' targets the deleted box. Item (2) ? DEPLOY_ALPHA.md still recommending split topology (line 29) and claiming LIVE (line 3) ? is the only still-actionable piece (covered by the doc-rot finding). DNS for app./api. still resolves via Cloudflare (dangling).
362 - why it matters:
363 - proposed fix: Re-scope or close-and-replace #97: the durable content is the decision

'single-origin is the default for SPA+Access' ? record it in the Cloud Run serving plan and DEPLOY_ALPHA.md, then close. The DNS/Access cleanup itself is Cloudflare-dashboard work.

364 - verifier notes: One scoping correction to the proposed fix: no "Cloud Run serving plan" doc exists in khelsutra (deploy/ holds only DEPLOY_ALPHA.md, GPU_VM_SERVING.md ? itself droplet-era "droplet kept as fallback" at DEPLOY_ALPHA.md:9 ? cloudflared.config.example.yml, droplet/, stage_from_bucket.sh). The durable single-origin decision should be recorded in DEPLOY_ALPHA.md (or the engine repo's Cloud Run docs, e.g. deploy/cloudrun/README.md there) rather than a not-yet-existing plan doc; alternatively the replacement issue itself can carry the decision. Also note issue #114 (CF Access allowlisted-email sign-in) is a second alpha-auth issue that references the same retired stack and likely needs the same re-scope sweep. Since closing the owner's own issue records/retires an owner decision, get a one-line owner ack in the close comment, but that does not gate the finding.

365 - verifier reasoning: (1) Issue exists and is stale-open: `gh issue view 97` (avidullu/khelsutra) ? state OPEN, createdAt 2026-06-22, updatedAt 2026-06-22 ? untouched for 10 days. Its body's "Fix applied (2026-06-22) ? single-origin" describes the engine-served SPA on `api.khelsutra.guru` through the droplet tunnel, and its remaining items (1) `CNAME app ? 55142398-...cfargotunnel.com` and (3) "refresh the engine's bundled SPA" both target that box. (2) That infrastructure is gone: memory MEMORY.md:63 "khelsutra site droplet parity ? RETIRED 2026-06-25 ... droplet deleted (\$0) ... Follow-ups: `api.` CNAME/CF-Access cleanup" and session-handoff.md:19 "the prior 'LIVE alpha on the droplet' ... is GONE ? alpha is paused pending the Cloud-Run move; api.khelsutra.guru CNAME/CF-Access need cleanup". (3) The doc anchor is as claimed: deploy/DEPLOY_ALPHA.md:3 still says "***Status:** `LIVE` ... **Topology:** split" and :29 still recommends "This runbook ships the **split** topology you chose". (4) Not already fixed: `git log --oneline -15 -- deploy/DEPLOY\_ALPHA.md` ends at 6e4dbaf (#113, droplet-era GPU-VM doc) and 7dc7de5 (#95); all commits since 2026-06-25 (f9eea14?dfc643f) are the storage-medium/SPA track ? none touch #97 or the runbook; no successor issue exists in `gh issue list` (the only topology tracker is #97). (5) DNS-dangling claim verified: nslookup via 1.1.1.1 ? both app. and api.khelsutra.guru resolve to Cloudflare proxy IPs (104.21.79.79 / 172.67.169.91) while the tunnel origin is deleted. Failure scenario is concrete: the next alpha session resuming from #97's checklist would reprint production `app.` DNS at a dead tunnel (530s) and try to refresh an SPA bundle on a nonexistent box. Not owner-gated: re-scoping/closing a stale tracker and recording an already-made owner decision (single-origin for SPA+Access, decided 2026-06-22 in the issue itself) is hygiene, not alpha go-live; the finder correctly leaves the CF-dashboard DNS/Access cleanup to the owner.

366

367 ### R1-36 [P2/S] doc-rot ? khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/
368 **NEXT_STEPS.md was refreshed 2026-07-01 but still advertises the retired droplet host as 'live + browsable'*

369 - anchor: `khelsutra/NEXT\_STEPS.md:12` . tracked in: ? . finder: khelsutra:code
370 - evidence: NEXT_STEPS.md:5 says 'updated 2026-07-01' yet :12-13 states the Node+SQLite host is 'live + browsable on the DO droplet ? http://168.144.126.176:8787 (systemd khelsutra-site)' ? six days after the droplet was decommissioned (commit 5ffb750, 2026-06-25). :61-64 still queues 'Droplet ? production ? TLS + a subdomain (vm.khelsutra.guru)' and 'Revoke the droplet SSH key (~/.ssh/khelsutra_droplet) when done' as if the box exists. deploy/droplet/README.md remains pre-baked for the same IP.

371 - why it matters:
372 - proposed fix: Refresh NEXT_STEPS.md 'Where we are': drop the live-droplet claim, retire the 'Droplet ? production' queue item (or re-scope it to 'any future VM' ? site/scripts/provision-vm.sh is already provider-agnostic), and keep the SSH-key revocation as an explicit still-actionable item. Mark deploy/droplet/README.md as a historical template.

373 - verifier notes: Sharper failure scenario: NEXT_STEPS.md is the repo's designated ramp-up doc ("Refresh at each checkpoint") and was refreshed 2026-07-01, so a new session trusts it as current state ? it then browses/SSHes a dead IP, or worse, starts the queued "Droplet ? production" TLS/Caddy work against a box that no longer exists. Meanwhile the ~/.ssh/khelsutra_droplet revocation (a real hygiene action that IS still outstanding) stays parked behind "when done" of work on nonexistent infrastructure. One correction to scope: the rot is wider than the two cited files ? docs/LOCAL_APPLIANCE_ARCHITECTURE.md:48 ("Droplet reality: 168.144.126.176... co-hosts the marketing site and the engine"), docs/DRY_RUN_CUJ.md:42, and deploy/DEPLOY_ALPHA.md:35 all still offer the droplet as a usable engine host; the fix PR should sweep those in the same pass (still S).

374 - verifier reasoning: Personally reproduced every cited element in the current working tree of C:/Users/avidu/Projects/khelsutra-guru/khelsutra (master == origin/master, clean). (1) NEXT_STEPS.md:5 reads "Where we are (updated 2026-07-01)" and NEXT_STEPS.md:12-13 state the Node+SQLite host is "live + browsable on the DO droplet ? http://168.144.126.176:8787** (systemd `khelsutra-site`)". (2) The droplet is genuinely retired: commit 5ffb750 (2026-06-25,

"chore(ci): retire the live-parity monitor ? droplet mirror decommissioned (#117)") says in its body "That droplet has been decommissioned (its data archived to gs://khelesutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/)... Cloudflare is now the single canonical host." (3) Not already fixed: the most recent commit touching NEXT_STEPS.md is 9223ba9 (2026-07-01, #139), whose diff bumps the "updated" date at line 5 while the 168.144.126.176 claim appears only as unchanged hunk context ? the stale text survives in HEAD/working tree. (4) NEXT_STEPS.md:61-62 still queue "Droplet ? production ? TLS + a subdomain (`vm.khelesutra.guru` via Caddy)" and :64 defers "Revoke the droplet SSH key (`~/.ssh/khelesutra\_droplet`) when done" as if the box exists. (5) deploy/droplet/README.md:4, :87, :100 remain pre-baked for 168.144.126.176 with a "Live-verified on the droplet (2026-06-22)" banner. (6) Proposed fix is sane: site/scripts/provision-vm.sh and teardown-vm.sh exist (commit 23e2104, "provider-portable VM provisioning + teardown"), so re-scoping the queue item to "any future VM" is grounded. Not owner-gated ? a pure docs refresh; only the SSH-key revocation itself is an owner action, and the fix correctly keeps it as an explicit open item rather than performing it.

375

376 ### R1-37 [P2/S] doc-rot ? khelesutra (avidullu/khelesutra) ? SPA web/ + marketing site/
377 **All 15 engine line-number contract anchors in the SPA API client are stale, and will drift further when engine PR #390 (main.py split) merges**
378 - anchor: `khelesutra/web/src/api/client.ts:87` · tracked in: engine PR #390 (the split that worsens it) · finder: khelesutra:code
379 - evidence: web/src/api/client.ts:87,105,110,156,223,228,238,247,256,288,297 and web/src/api/types.ts:2-3,33,143,172 cite '[main.py:260]' (process), ':303' (jobs), ':447' (videos), ':331' (query), ':307' (highlights), ':121' (config), '[uploads.py:64/89/123/147]'. Actual engine routes today: /api/process main.py:1355, /api/jobs main.py:1457, /api/videos main.py:521, /api/query main.py:1523, /api/highlights main.py:1492, /api/config main.py:379; uploads.py:70/111/157/188. Every anchor is wrong; these comments are the repo's 'verified contract' convention, so wrong anchors actively mislead the next contract audit. Engine PR #390 (extract orchestration.py from main.py) is still OPEN (updated 2026-06-26) and will invalidate whatever numbers remain.

380 - why it matters:

381 - proposed fix: Replace hard line numbers with stable route anchors ? e.g.

'backend/main.py @app.post("/api/process")' / 'backend/api/uploads.py @router.post("/api/upload/init")' ? in one mechanical pass over client.ts + types.ts. Do it after (or regardless of) #390 so the anchors survive the god-module split.

382 - verifier notes: Finding is slightly UNDERSTATED: counting the three anchors packed into types.ts:3 separately, it is 17 anchors, and the finder missed that the types.ts:2 anchor 'backend/infrastructure/database.py:124-135 (play_segments)' is ALSO stale ? the play_segments CREATE TABLE now sits at database.py:204 (lines 124-135 are a migration if-chain). Include database.py in the mechanical fix pass. Also client.ts:238's '[main.py:362]' points at the CompileRequest model region, not the min_shot_count enforcement ? same fix applies (anchor to the /api/compile decorator or the config key name).

383 - verifier reasoning: Personally reproduced in current trees (both repos synced with origin/master). SPA anchors read directly: web/src/api/client.ts:87 '[main.py:260]', :105 '[main.py:303]', :110 '[main.py:447]', :156 '[main.py:331]', :223 '[main.py:307]', :228 '[main.py:121]', :238 '[main.py:362]', :247/:256/:288/:297 '[uploads.py:64/89/123/147]'; web/src/api/types.ts:2-3 (database.py:124-135, main.py:331-396/:307-328/:121), :33, :143, :172. Engine actuals grepped from badminton-highlight-indexer@e410136: /api/config main.py:379, /api/videos :521, /api/process :1355, /api/jobs :1457, /api/highlights :1492, /api/query :1523, /api/compile :1594; uploads.py :70/:111/:157/:188 ? every anchor wrong, exactly as claimed. Read the cited lines: main.py:258-262 = CORS middleware + '/' HTML route, :303-310 = legacy load_config wrapper, :118-124 = _finalize_reingest. Not fixed: git log on client.ts/types.ts tops at f9eea14 (#140), no anchor refresh. PR #390 verified OPEN via gh (updated 2026-06-26, files backend/main.py + backend/orchestration.py) ? merge shifts numbers further. Severity honest: types.ts:1-3 declares these the 'verified contract' anchors, and the /api/videos contract demonstrably drifted (engine #412/#413 ? web #140 resilient readers), so a stale anchor can falsely 'verify' a live contract. Fix (route-decorator anchors, comment-only, 2 files) sane, effort S. Not owner-gated: comment hygiene, no go-live/corpus/spend/legal surface.

384

385 ### R1-38 [P2/M] test-gap ? khelesutra (avidullu/khelesutra) ? SPA web/ + marketing site/

386 **SPA e2e CUJ never runs in CI and the mock engine is hand-synced ? the mock-vs-real drift bug-class already shipped once**

387 - anchor: `khelesutra/.github/workflows/test.yml:31` · tracked in: e2e/mock-engine.mjs:5 ('later drive CI CUJ replays') · finder: khelesutra:code

388 - evidence: .github/workflows/test.yml:31-49: the web job runs only typecheck + test:coverage + build ? the Playwright CUJ specs (web/e2e/ingest.pw.mjs, annotate.pw.mjs, store-

medium.pw.mjs against e2e/mock-engine.mjs) are never executed in CI (the workflow's only Playwright job is the site/ layout suite, :51-75). e2e/mock-engine.mjs:5 admits this is deferred: 'used to record the e2e screencast and (later) drive CI CUJ replays'. The cost of hand-syncing the mock is documented in web/src/lib/video.ts:4-6: 'the bug this guards against actually shipped once: the real engine returned id/no-name while the e2e mock faked both, so a green UI was broken live.'

389 - why it matters:

390 - proposed fix: Add a fourth CI job (mirroring the existing layout job's Playwright setup) that builds web/ and runs the e2e CUJ specs against mock-engine.mjs; separately add a thin contract check pinning the mock's response shapes to web/src/api/types.ts (or an engine-side snapshot of _video_summary/capabilities) so mock-vs-real drift fails a build instead of shipping.

391 - verifier notes: Path correction: tracked_in/evidence path is web/e2e/mock-engine.mjs:5 (no repo-root e2e/ exists); the finding's file field (.github/workflows/test.yml:31) is correct. Fix scoping addendum: the CI job needs a CI profile for playwright.config.mjs ? outputDir defaults to a sibling non-git khelsutra-screencast/ dir (line 25, override via SCREENCAST_DIR) and slowMo:350 + always-on video recording (lines 29-30) are screencast-oriented and should be disabled in CI. Severity P2 is honest: the exact mock-vs-real drift bug-class shipped once (video.ts:5-6), but defense-in-depth readers in web/src/lib/video.ts mitigate the known instance and the CUJ still runs manually per screencast checkpoint. Not owner-gated ? pure CI/test infrastructure.

392 - verifier reasoning: khelsutra/.github/workflows/test.yml (only file in .github/workflows/): web job lines 31-49 runs only typecheck + test:coverage + build; the sole Playwright job is `layout` lines 51-75 with working-directory: site (line 59). web/package.json:14 defines an `e2e` script never invoked by CI. web/e2e/ contains ingest.pw.mjs, annotate.pw.mjs, store-medium.pw.mjs driven against mock-engine.mjs via playwright.config.mjs:32-38 (webServer boots the mock; line 35 `reuseExistingServer: !process.env.CI` shows CI was anticipated but never wired). web/e2e/mock-engine.mjs:5 confirms the deferral: "(later) drive CI CUJ replays". web/src/lib/video.ts:4-6 confirms the shipped drift bug verbatim. No contract check pins the mock to web/src/api/types.ts (repo grep for mock-engine hits only docs + config; docs/STUDIO_MEDIA_LIFECYCLE.md:156 shows hand-coordination of the P3 shape). git log for test.yml: latest commit 6f17dc0 is action-version bumps only ? not fixed at HEAD (f9eea14).

393

394 ### R1-39 [P2/S] hygiene ? sports-data-collector

6. user (2026-07-02T04:50:04.945Z)

NEXT_STEPS.md
 README.md
 deploy
 docs
 locales
 node_modules
 site
 web
 C:/Users/avidu/Projects/khelsutra-guru/khelsutra/deploy:
 DEPLOY_ALPHA.md
 GPU_VM_SERVING.md
 cloudflared.config.example.yml
 droplet
 stage_from_bucket.sh

C:/Users/avidu/Projects/khelsutra-guru/khelsutra/deploy/droplet:
 README.md
 config.json.example
 engine.env.example
 khelsutra-engine.service

7. user (2026-07-02T04:50:06.299Z)

f9eea14 web: resilient video-summary readers + show fps in the library (#140)
 9223ba9 docs: refresh khelsutra docs for today's storage-track work (P2-P5) (#139)
 fb2f561 web: persistent "FFmpeg not detected" banner (hard-blocker warning) (#138)
 8667dbf P5: SPA storage-medium picker (store a video into your medium) (#137)

```
a04a21b docs(tracker): inline the D14 decisions into the P2-P5 rows (#136)
commit 5ffb7504a80550b5a5bc0e55b901b7051eddb0aa
Author:      Avi <avi.dullu@gmail.com>
AuthorDate:  Thu Jun 25 16:50:58 2026 +0530
Commit:      GitHub <noreply@github.com>
CommitDate:  Thu Jun 25 16:50:58 2026 +0530
```

chore(ci): retire the live-parity monitor ? droplet mirror decommissioned (#117)

The scheduled `live-parity` workflow compared the Cloudflare-served site (canonical, auto-deployed from git) against a second live copy on the DigitalOcean droplet `168.144.126.176:8787`. That droplet has been decommissioned (its data archived to `gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/`), so the monitor has nothing to compare against and fails every night.

Remove the workflow and its only consumer, `parity-check.sh`. Cloudflare is now the single canonical host. `test.yml`'s ops-script check globs `scripts/\*.sh`, so dropping the script doesn't affect it.

Co-authored-by: Claude Opus 4.8 <noreply@anthropic.com>

```
.github/workflows/parity.yml | 28 -----
site/scripts/parity-check.sh | 22 -----
2 files changed, 50 deletions(-)
```

8. assistant (2026-07-02T04:50:15.631Z)

Retirement facts verified via git log (commit 5ffb750, PR #117). No CLAUDE.md at the khelsutra repo root ? let me double-check, then create the isolated worktree.

9. user (2026-07-02T04:50:16.814Z)

No files found

10. user (2026-07-02T04:50:19.445Z)

Preparing worktree (new branch 'docs/retire-droplet-doc-rot')
branch 'docs/retire-droplet-doc-rot' set up to track 'origin/master'.
HEAD is now at f9eea14 web: resilient video-summary readers + show fps in the library (#140)

11. assistant (2026-07-02T04:50:27.950Z)

Worktree created. Now let me read all target files.

12. user (2026-07-02T04:50:29.480Z)

```
1      # Deploy KhelSutra Studio for alpha ? a shareable page, no SSH
2
3      > **Status:** `LIVE` (deploy live-verified on the droplet 2026-06-22 ?
`deploy/droplet/README.md`; only a rotated Gemini key + a browser OTP login remain to exercise
the upload?reel CUJ) . **Topology:** split ? SPA on Cloudflare Pages + engine on an `api.`
cloudflared tunnel, Cloudflare Access over both (the engine's `HOSTING_PLAN.md` $2 alpha plan).
. **Cost:** ~$0/mo standing (Pages + Tunnel + Access are free ?50 users) + Gemini per-run
(~$0.05/match).
4
5      This turns the local-only flow (run engine + SPA on your box, drive it yourself) into a
**page you can share with alpha testers** ? they open `app.khelsutra.guru`, sign in (email
allowlist), point at a video, and get a reel. No SSH, no home IP exposed, no inbound ports.
6
7      > **Outgrown the droplet?** To move the live engine onto a **GCP L4 GPU VM** (more
CPU/RAM for ffmpeg
8      > + optional WASB GPU detection) and cut the tunnel over to it, see
**[`GPU_VM_SERVING.md`](GPU_VM_SERVING.md)**
```

```

9      > ? same tunnel + Access, no DNS change, droplet kept as fallback (~$500?600/mo while
the VM runs).
10
11     ## 0. The shape
12
13     ```
14     tester browser
15     ? https
16     ?
17     Cloudflare edge (free): DNS · TLS · Access (email-OTP allowlist) in front of BOTH
hostnames
18     ??? app.khelsutra.guru ? Cloudflare Pages          (the web/ SPA; built with
VITE_API_BASE)
19     ??? api.khelsutra.guru ? cloudflared Tunnel (outbound-only) ?? 127.0.0.1:8000 on
YOUR box
20                                     ? FastAPI engine
21                                     ? (Gemini
segmenter; GPU optional)
22                                     ? async jobs,
SQLite, output/ on disk
23     ```
24
25     - **No inbound ports.** `cloudflared` dials out; the engine stays bound to
`127.0.0.1:8000`. The box's home/public IP is never exposed.
26     - **Auth at the edge.** Cloudflare Access gates both hostnames; the engine *also*
verifies the `Cf-Access` JWT in-app (defence in depth ? `security.edge_auth_enabled`), so a
direct hit to the tunnel can't spoof identity.
27     - **Cross-origin, on purpose.** `app.` (SPA) and `api.` (engine) are different origins ?
the SPA is built with `VITE_API_BASE=https://api.khelsutra.guru/api` (PR #64) and the engine
locks CORS to the SPA origin via `RALLY_CORS_ALLOW_ORIGINS` (no engine code change ? it's
already env-driven).
28
29     > **Alternative (single-origin):** if you'd rather run **one box** that serves the SPA
*and* the API behind one origin (Caddy reverse-proxy, no Pages, no cross-origin), see
`LOCAL_APPLIANCE_ARCHITECTURE.md` D1. This runbook ships the **split** topology you chose
(stable Pages-hosted page + a thin API tunnel).
30
31     ---
32
33     ## 1. Prerequisites
34
35     - A **box** to run the engine: the CPU droplet `168.144.126.176` (Gemini-only ? no GPU;
fine for alpha) **or** your Windows+WSL GPU box (real WASB). `ffmpeg` + Python 3.8+ installed;
the `badminton-highlight-indexer` engine checked out.
36     - A **Cloudflare account** with the `khelsutra.guru` zone (you already run the marketing
site there).
37     - `cloudflared` installed on the box (`https://developers.cloudflare.com/cloudflare-
one/connections/connect-networks/downloads/`).
38     - A **fresh** `GEMINI_API_KEY` (rotate the chat-shared one ? it uploads downsized chunks
to Google; this is a paid + cloud action).
39
40     ---
41
42     ## 2. Engine (the `api.` side)
43
44     ### 2.1 Install (with the auth extra)
45     ```bash
46     cd badminton-highlight-indexer
47     pip install -e .[auth]          # PyJWT[crypto] ? REQUIRED when edge auth is on, else
every authenticated request is rejected 401
48     ```
49
50     ### 2.2 The safe-exposed config
51     Create / edit `config.json` on the box. **Where it must live:** the engine reads
`$RALLY_APP_HOME/config.json` if `RALLY_APP_HOME` is set, else `./config.json` relative to *the

```

```

directory you launch the server from* ? **not** necessarily the repo root. If the engine reads a
different (or absent) file it silently falls back to all-defaults, which means **edge auth is
OFF**. To make writer and reader agree by construction, write it with `python -m backend.main
--setup` (it targets the exact path the server reads) and edit that file:
52     ``jsonc
53     {
54         "security": {
55             "edge_auth_enabled": true,                // turn the CF-Access JWT verify ON
56             "allowed_video_root": "/srv/khelsutra/incoming", // the path-jail ? REQUIRED when
exposed
57             "upload_dir": "/srv/khelsutra/incoming/uploads", // MUST sit UNDER
allowed_video_root (see §7)
58             "cf_team_domain": "https://<your-team>.cloudflareaccess.com",
59             "cf_access_aud": "<the CF Access application AUD>" // filled in §4
60         },
61         "indexing": {
62             "default_segmenter": "gemini" // ? CPU box MUST set this ? the shipped default
`yolo_hybrid` needs torch/GPU, and the SPA sends no segmenter, so every job fails "No module
named 'torch'" without it
63         }
64     }
65     ``
66     And export the env (a `.env` or your shell):
67     ``bash
68     export GEMINI_API_KEY="<your-rotated-key>"
69     export RALLY_CORS_ALLOW_ORIGINS="https://app.khelsutra.guru" # lock CORS to the SPA
origin
70     export RALLY_ALLOWED_HOSTS="api.khelsutra.guru,localhost,127.0.0.1" # REQUIRED ? see ?
below; without it EVERY tunneled request 400s (loopback added so on-box /api/health checks still
pass ? setting this var REPLACES the loopback-only default)
71     export RALLY_BIND_HOST="127.0.0.1" # keep the loopback bind
(cloudflared dials 127.0.0.1:8000)
72     ``
73
74     > **? Two env vars you MUST set, or the tunnel is dead-on-arrival.** With
`edge_auth_enabled: true`, the engine (a) flips its bind to `0.0.0.0` and (b) keeps the DNS-
rebinding **Host-header guard** at its loopback-only default ? so it boots "healthy" but then
**400's every request `cloudflared` forwards with `Host: api.khelsutra.guru` (including the
§6 `/api/health` check). The boot only *warns* about this (`EXPOSED_HOST_ALLOWLIST_LOOPBACK`),
it does not stop. Set **`RALLY_ALLOWED_HOSTS="api.khelsutra.guru,localhost,127.0.0.1"` (the
load-bearing fix ? the Host allowlist is independent of the bind; the loopback entries keep the
on-box `/api/health` check passing, since setting this var REPLACES the loopback-only default)
and **`RALLY_BIND_HOST="127.0.0.1"` (so the bind stays loopback as §0/§6 claim; `cloudflared`
dials loopback anyway). *Verified against engine `origin/master`: with the allowlist unset, `GET
/api/health` with `Host: api.khelsutra.guru` ? `400`; with it set ? `200`.*
75
76     > **The engine refuses to start unsafe.** The boot posture (engine PR #294) **fails
fast** if `edge_auth_enabled` is on but the `allowed_video_root` jail or
`cf_team_domain`/`cf_access_aud` is missing, and **warns** if `PyJWT` isn't installed. So a
half-configured exposure can't silently come up ? fix what it prints and restart.
77
78     ### 2.3 Run
79     ``bash
80     python -m backend.main --server # binds 127.0.0.1:8000 ? kept loopback by
RALLY_BIND_HOST (§2.2); with edge_auth ON and that var unset it would bind 0.0.0.0 instead
81     ``
82     Keep it alive with a process manager (`systemd` on the droplet, mirroring
`site/scripts/provision-vm.sh`'s unit; or `pm2`/`nssm` on Windows).
83
84     ---
85
86     ## 3. The SPA on Cloudflare Pages (the `app.` side)
87
88     The SPA ships as a **static Pages site**, git-connected like the marketing site (auto-
deploy on merge to `master`).

```

```

89
90 1. **Cloudflare dashboard ? Workers & Pages ? Create ? Pages ? Connect to Git** ? pick
the `khelsutra` repo.
91 2. **Build settings:**
92   - **Root directory:** `web`
93   - **Build command:** `npm run build`
94   - **Build output directory:** `web/dist`
95   - **Environment variable:** `VITE_API_BASE = https://api.khelsutra.guru/api` ? this
is what points the SPA at the engine tunnel (PR #64).
96 3. **Custom domain:** add `app.khelsutra.guru` to the Pages project.
97
98 > Every push to `master` now rebuilds + redeploys the SPA. To preview a change first,
Pages builds a per-PR preview URL.
99
100 ---
101
102 ## 4. Cloudflare Access (gate BOTH hostnames)
103
104 1. **Zero Trust ? Access ? Applications ? Add ? Self-hosted.**
105 2. **Application domain:** add **both** `app.khelsutra.guru` **and**
`api.khelsutra.guru` to the same application (one app ? one shared session cookie, so a tester
signs in once and both the SPA and its API calls are authorized).
106 3. **Policy:** Action *Allow*, rule *Emails* ? your alpha testers' addresses (free ? 50
users ? fits 10?25 testers). Use one-time-PIN (email OTP) so testers need no Cloudflare account.
107 4. **Copy the application `AUD` tag** ? paste into the engine's `config.json`
`security.cf_access_aud` (§2.2), and set `cf_team_domain` to `https://<your-
team>.cloudflareaccess.com`. Restart the engine.
108 5. **CORS preflight ?:** the browser sends an unauthenticated `OPTIONS` preflight to
`api.` for the cross-origin POSTs. In the Access application's **Settings ? enable ?Bypass
options requests to CORS preflight?*** (or add an explicit OPTIONS bypass), else the preflight is
redirected to the login page and every API call fails. The engine's CORS middleware
(`allow_credentials=false`, no cookies) then answers the preflight.
109
110 ---
111
112 ## 5. The tunnel (engine ? Cloudflare)
113
114 ```bash
115 cloudflared tunnel login # browser auth to your zone
116 cloudflared tunnel create khelsutra-studio # note the <TUNNEL_UUID>
117 # copy deploy/cloudflared.config.example.yml ? /etc/cloudflared/config.yml, fill
<TUNNEL_UUID>
118 cloudflared tunnel route dns khelsutra-studio api.khelsutra.guru
119 cloudflared tunnel run khelsutra-studio # or install as a service:
`cloudflared service install`
120 ```
121 See `deploy/cloudflared.config.example.yml` for the ingress (only `api.khelsutra.guru` ?
http://127.0.0.1:8000).
122
123 ---
124
125 ## 6. Verify (the posture test + the CUJ)
126
127 **Exposure posture (must hold ? this is what makes it safe to share):**
128 - From OUTSIDE the box: `curl --max-time 5 http://<box-public-ip>:8000/api/health` ?
**refused / timeout** (the engine is `127.0.0.1`-bound; no inbound port). ? The box is not
directly reachable.
129 - `curl -i https://api.khelsutra.guru/api/health` **without** a CF session ? a **302 to
the Access login** (not a 200). ? The gate is in front.
130 - Open `https://app.khelsutra.guru` in a browser ? CF Access login ? after sign-in, the
SPA loads and `/api/health` returns 200 (same session cookie). ?
131
132 **The shareable CUJ (what a tester does):** open `app.khelsutra.guru` ? sign in ?
**upload a video from their device** (the SPA chunked-upload lands it in `upload_dir`, inside
the jail) ? **Process** ? watch progress ? pick rallies ? **Build** ? **download the reel**. All

```

```

from the page. *(Pasting a `gs://` URI does NOT work on this box ? see §7: the
`allowed_video_root` jail rejects every non-local URI. Browser upload is the alpha path.)*
133
134 ---
135
136 ## 7. Honest caveats / deferred
137
138 - **`upload_dir` must sit UNDER `allowed_video_root`.** If you enable browser upload, an
upload landing outside the jail is rejected by `/api/process` (a functional break, not a hole).
Keep `upload_dir` inside the jail (§2.2).
139 - **`gs://` / `s3://` / `gdrive://` URIs are REJECTED on the exposed box.** Setting
`allowed_video_root` (required to expose, §2.2) path-jails `/api/process` to *local* paths under
that root; the engine resolves a `gs://` URI to a local path that escapes the jail ? `400`
(pinned by engine `tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri`).
Cloud-URI ingest works **only** when `allowed_video_root` is unset (the local single-user
posture ? the opposite of this runbook). For alpha, the ingest path is **browser upload** (it
lands inside the jail). *(Scheme-exempting recognized buckets from the jail would be an engine
change, not a config flip.)*
140 - **You MUST set `edge_auth_enabled=true` to expose.** The boot posture (and the in-app
JWT verify) only engage then. Do not put the engine behind the tunnel with edge auth off.
141 - **Cross-origin reel download:** the reel streams from `api.` with `Content-
Disposition: attachment`, so it downloads even cross-origin ? but verify the filename in your
browser during the CUJ (cross-origin <a download>` filename hints are ignored; the header is
the authority).
142 - **Upload size:** Cloudflare's free proxy caps request bodies at **100 MB** ? large
matches upload as ~95 MB chunks (the SPA chunked-upload path honours the engine's `max_part_mb`,
default **95** ; lower `security.max_part_mb` to shrink them). `gs://` bucket pre-staging is
**not** an option on this box (see the jail caveat above).
143 - **Box offline = engine offline.** The Pages SPA stays up and shows a friendly "engine
offline" banner (`/api/health` ping); processing resumes when the box is back. Beta removes this
with a queue (`HOSTING_PLAN.md` §3).
144 - **Gemini = paid + uploads footage to Google.** Keep the key rotated; use it only for
clips you're OK sending.
145
146 ---
147
148 ## 8. Teardown
149 `cloudflared tunnel cleanup khelsutra-studio` + delete the tunnel; remove the Pages
custom domain; disable the Access application; stop the engine service. The box's `output/` +
`uploads/` retain originals ? sweep them per your retention policy before any wider exposure.
150
151 ---
152 ### Changelog
153 - 2026-06-22 ? **correctness pass** (adversarial verify against engine `origin/master` +
a hermetic repro). Fixed two would-break-the-deploy gaps: (1) **`RALLY_ALLOWED_HOSTS` was
missing** ? with `edge_auth` on, the Host-header guard 400s every tunneled request until it's
set (added it + `RALLY_BIND_HOST` to §2.2; repro confirmed 400?200); (2) **`gs://` ingest is
rejected** by the `allowed_video_root` jail on the exposed box (dropped the `gs://` promise from
§6/§7; alpha path = browser upload). Minor: 401-not-500 wording (§2.1), `config.json`
location/`--setup` note (§2.2), bind comment (§2.3), ~95 MB chunk size (§7).
154 - 2026-06-21 ? created. Split-topology alpha deploy kit (cloudflared + Pages + CF
Access) consuming the engine exposure-safety boot posture (engine #294) + the SPA
`VITE_API_BASE` (#64). Pairs with `LOCAL_APPLIANCE_ARCHITECTURE.md` (single-origin alternative)
+ the engine `HOSTING_PLAN.md` §2.
155

```

13. user (2026-07-02T04:50:30.258Z)

```

1 # Next steps (live) ? KhelSutra brand / site
2
3 Prioritized "what to do next" for the **`khelsutra` product repo**. Refresh at each
checkpoint.
4
5 **Where we are (updated 2026-07-01):** `khelsutra.guru` is **LIVE and fully shipped** ?
landing page + `/capture`

```

```

6      checklist + OG image + a **segmented Request-Access form** (academy/individual + GPU +
storage + free
7      text). On **Cloudflare** (Worker `hidden-king-khelsutra` + D1 `khelsutra-access`): the
form persists to
8      D1 **and emails a summary to the owner on every registration** (Email Routing send
binding ? verified
9      live). **Git auto-deploy is connected** (Workers Builds, root `site/`): merge to
`master` ? deploys;
10     **CI gates every PR** ? three jobs: the full site `npm test` (Worker + VM host + assets
+ UX), the
11     `web/` SPA build + coverage floor, and a real-browser layout check. The same form also
runs on a
12     dependency-free **Node + SQLite** host (`site/server.js`), **live + browsable on the DO
droplet ?
13     http://168.144.126.176:8787 ** (systemd `khelsutra-site`). Shipped across ~90 merged PRs
(the form,
14     6-language localization, the per-rally MVP, droplet portability). Docs:
15     `docs/REQUEST_ACCESS_FORM.md`, `docs/PORTABILITY.md`.
16
17     ## Next product threads (owner ideas, 2026-06-20)
18     - **? DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22; human gates cleared):** the
product is
19     non-tech-friendly and **fully localized to all 6 languages** (en/kn/hi/es/da/id)
across both surfaces
20     (`web/` SPA + `site/` marketing) ? including the home page (hero, all sections, the
`#join` access form,
21     footer), the **Privacy Policy** page, **per-page `<title>s**, and **es/da/id academy
flyers** (with
22     localized SPA screenshots). The responsive header is fixed and a **headless-Chromium
CI suite** guards
23     layout, cross-locale render (live DOM == catalog), and **accessibility (axe-core WCAG
2.1 AA)**. Full
24     record: **`docs/archives/FRIENDLY_AND_MULTILINGUAL.md`**. Post-archive PRs: #86 (own-
model) . #87 (es/da/id
25     site) . #105 (form+footer + render guard) . #106 (Privacy page) . #107 (a11y fixes +
axe guard + page
26     titles) . #109 (flyers). **Owner gates CLEARED:** native-speaker review of the AI-
drafted catalogs ? +
27     real-device font-glyph eyeball ?. **Remaining (small/owner):**
28     - **Delta native-review** of the strings ADDED after the owner's sign-off
(#105/#106/#107/#109 ? the
29     form/footer/privacy/title/flyer copy is AI-drafted): a small top-up review before a
wide public launch.
30     - **B7 (deferred, not a v1 blocker):** engine backend customer-facing l10n (report
`customer_` +
31     rejection reasons) ? cross-repo Python (`badminton-highlight-indexer`), owner-gated
**D-BACKEND-DEPTH**.
32     - **Marketing video on the site** ? embed a short walkthrough of the record ? index ?
reel process,
33     themed for **serious enthusiasts**, on the landing page. ? **A real-UI CUJ screencast
now exists**
34     (Playwright recording of the per-rally SPA driven against real GPU-run data ?
`promo_cuj.mp4`/`.gif`
35     in `gs://khelsutra-rally-corpus/promo/`). Decide: use it as the marketing clip (?
shows real court
36     footage + a dev "P5" badge ? clean up before public embed), and/or produce a polished
Grok/Gemini cut.
37     - **Interactive per-rally selection** (big product feature) ? ? **MVP / M0 COMPLETE**
(the full
38     pick ? query ? select ? compile ? download loop on existing engine endpoints). Lives
in the alpha SPA
39     (`web/`). **? Tracked design + P-tracker (source of truth):
`docs/PER_RALLY_SELECTION.md` (IN PROGRESS;
40     owner-approved, D8?D11 locked). ** **Done: P0 design (#11), P1 scaffold (#14), P2
grid+hook (#21), P3

```

```

41     `parseQuery()` query bar (#28), P4 SelectionFooter + `min_shot_count` warning (#30),
P5 real load +
42     Build?compile?preview?download (#31).** Verified e2e vs a stub engine + a CUJ replay
against the real
43     L4-run rallies + reel. **NEXT (owner-gated, engine repo): M1?M5** ? `GET /api/health`
+ `download_url`
44     (P6), RESTful `~/rallies` alias (P7), multi-video home + auth (P8), per-rally
preview/timeline (P9),
45     corpus correction loop (P10), localized NL (M5). **Optional:** a fresh live
`/api/process` GPU
46     inference once the engine session's M2 training frees the single GPU.
47
48     ## Queued (site polish)
49     - **Always-Use-HTTPS + HSTS** (Cloudflare SSL/TLS) ? closes the plain-HTTP flag.
50     - **Cloudflare Web Analytics (owner-gated; FREE)** ? code side DONE (the dead
placeholder beacon was
51     removed from `site/public/index.html`; analytics is now via **Automatic Setup**, no
in-source token).
52     **Owner step (needs CF login):** the site is a Cloudflare **Worker** (not Pages) on
the proxied
53     `khelesutra.guru` zone ? enable via **Analytics & Logs ? Web Analytics ? Add a site ?
`khelesutra.guru`**
54     (Automatic Setup is default-on for proxied hostnames; edge-injects the beacon on every
page). Then
55     filter the view to `hostname = khelesutra.guru` to exclude `*.pages.dev`/preview noise,
and decide
56     separately whether `app.khelesutra.guru` (the SPA) needs its own enablement (history-
router only).
57     **Verify (two gotchas):** the Worker's asset responses must NOT send `Cache-Control:
?, no-transform`
58     (Cloudflare can't rewrite that ? no injection), and after a deploy the rendered HTML
should contain the
59     edge-injected beacon. **DONE WHEN:** the Web Analytics dashboard shows incrementing
page views for
60     `khelesutra.guru`.
61     - **Droplet ? production** ? TLS + a subdomain (`vm.khelesutra.guru` via Caddy) if it
should be a real
62     standby; add **VM-path SMTP** so the Node host also emails (currently `noopNotify`).
63     - **DMARC** TXT for deliverability; delete any orphan `khelesutra-site` Worker; clean CF
D1 test rows
64     (`deploy-test@`, `email-test@`). Revoke the droplet SSH key
(`~/ssh/khelesutra_droplet`) when done.
65
66     ## Bigger arcs (design-first)
67     - **? Studio media lifecycle (ingest ? store ? annotate ? train)** ? **Tracked project:
`docs/STUDIO_MEDIA_LIFECYCLE.md`
68     (IN PROGRESS ? storage half COMPLETE 2026-07-01).** The seamless flow: add a video ?
store it in a user-chosen medium
69     (S3/GCS/Drive/local) ? free-space checked ? annotate in-UI ? contribute to the
segmenter. **SHIPPED 2026-07-01:**
70     free-space guards (vault + engine, P1/P2), the `/api/capabilities` storage block (P3),
`POST /api/assets`
71     store-into-medium (P4), the SPA medium picker (P5), the "my videos" library (P6), and
in-UI annotation (P7?P9).
72     **Remaining = owner-gated only:** P10 (corpus promotion) + P11 (train/re-eval ?F1) +
the `L` beta follow-ups.
73     §7 tracker is the source of truth.
74     - **Single-box appliance (Khelesutra Studio)** ? **? Tracked design:
`docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (DRAFT, owner-gated).**
75     The `web/` SPA drives the full **ingest (Drive / bucket URI / local upload) ? index ?
pick rallies ? reel ? download**
76     loop on ONE box, behind an **edge auth gate** (engine stays `127.0.0.1`, CORS `` is
moot). It is the single-box
77     profile of the engine's decoupled `StorageBackend` + `backend.worker` architecture.
MVP runs on the CPU droplet via

```

```

78      **Gemini/stub** ; **real local-GPU detection needs a GPU host** (DO droplet / GCE VM /
Cloud Run+GPU ? owner priority
79      ASAP, runs parallel to the UI MVP). Per-rally selection = one screen. Owner decisions
pending in §10 (edge auth,
80      retention, cost ceiling, GPU host, P5 bridge).
81      - **Alpha SPA** in `web/` (Vite + React + TS + Tailwind) ? hosts the operator console +
the per-rally selection UI above.
82      - **BYO-storage + autosync** ? connect Drive/OneDrive/GCS/S3, watch the folder, auto-
prepare reels
83      (engine `PLATFORM_ARCHITECTURE.md §3`). The form's GPU/storage answers are the demand
signal.
84      - **Control plane** (`control-plane/`) at beta.
85
86      ## Org & repos
87      - ? **DONE (2026-06-20): proprietary repos consolidated into the `Khelsutra` org
(private).** Transferred
88      `badminton-highlight-indexer` (engine), `sports-data-collector`, `sports-obsreport` ?
`Khelsutra/` (stay private; no
89      open PRs/forks; GitHub redirects git ops so active sessions kept working; local
remotes repointed + verified).
90      `rally-annotator` stays in the org **PUBLIC** (OSS MIT plugin the site links to ? do
not make private). Owner is org
91      admin (full access) and can **invite a contributor** (org member / per-repo outside
collaborator) anytime.
92      - ? **`khelsutra` (this site repo) ? owner step pending.** A transfer breaks the
**Cloudflare Workers-Builds** git
93      connection (auto-deploy) until the Cloudflare GitHub App is installed on the
`Khelsutra` org and the Worker
94      reconnected (dashboard). When ready: install the CF app on the org ? `gh api
repos/avidullu/khelsutra/transfer -f
95      new_owner=Khelsutra` ? reconnect the Worker to `Khelsutra/khelsutra` ? verify a
deploy.
96      - ? **Engine CI under the org** ? verify the engine's GitHub Actions run on the next PR
(free orgs default Actions-on;
97      enable in org settings if a run doesn't start).
98
99      ## Convention
100     Substantial work gets a tracked doc in `docs/`; one PR per item off `origin/master`;
merge to `master`
101     auto-deploys (CI must be green).
102

```

14. user (2026-07-02T04:50:36.022Z)

```

1      # Move KhelSutra Studio onto an L4 GPU VM (off the droplet)
2
3      Self-serve runbook to run the live engine (behind `api./app.khelsutra.guru`) on a **GCP
L4 GPU VM**
4      instead of the CPU droplet, and **cut the Cloudflare tunnel over** so traffic lands on
the VM. Lets
5      you switch detection between **Gemini** (cloud API, proven) and **WASB native** (uses
the L4 GPU).
6
7      This extends [`DEPLOY_ALPHA.md`](DEPLOY_ALPHA.md) (which deployed the same engine on the
droplet) ? it
8      reuses the droplet config templates in [`deploy/droplet/`](droplet/) and the GPU-VM
scripts in the
9      **engine repo** (`badminton-highlight-indexer`: `deploy/gcp/`).
10
11     > **These are manual steps today.** Automating them with a one-shot **startup-script**
you feed into
12     > the GCP console (`create_gpu_vm.sh` already injects a metadata startup-script ? today
it only installs
13     > the Ops Agent) is backlogged: engine repo `docs/BACKLOG.md` ? *Batch / declarative
orchestration*. The

```

```

14     > design hinge there is keeping the secrets (Gemini key / Access AUD / tunnel token) out
of plaintext
15     > metadata.
16
17     > **Cost ?** A `g2-standard-4` (1xL4) bills **~$0.70?0.85/hr ? ~$500?600/month** while
it runs. You
18     > keep it up, so this is an ongoing cost. **`bash deploy/gcp/teardown.sh delete` is the
only way to
19     > stop billing** (a *stopped* VM still bills its disk). For bursty/intermittent load, a
scale-to-zero
20     > **Cloud Run GPU *service** would be far cheaper (~$0 idle) ? that's the backlogged
"M7" and a bigger
21     > build; this VM is the fast path that works today.
22     >
23     > **Cutover is safe + reversible.** Bring the VM up and **verify it serving** before
touching the
24     > tunnel; the droplet stays as a hot fallback. The tunnel is **token-managed** (ingress
lives in the
25     > Cloudflare dashboard, not on the box), so moving it is just "run the same connector on
the VM, then
26     > stop it on the droplet" ? **no DNS change, no new Cloudflare token, no CF-Access
change.**
27
28     ---
29
30     ## 0. Prerequisites (gather these once)
31
32     - **gcloud** authenticated as the project owner (`gcloud auth login`) on the machine you
run this from.
33     - **GPU quota** `GPUS_ALL_REGIONS ? 1` and the SA Ops-Agent roles ? see the engine repo
34     `deploy/gcp/README.md` $2. (You already have L4 quota=1 from the nightly runs.)
35     - **The two secrets**, which you already have from the droplet deploy:
36     - the **rotated Gemini API key** (Drive: `gemini.api_key.txt`) ? only needed for the
Gemini path;
37     - the **CF-Access AUD** + team domain ? already in the droplet's
`/srv/khelsutra/config.json`
38     (`security.cf_access_aud` / `cf_team_domain`); reuse the same values.
39     - the **tunnel token** ? in the droplet's cloudflared unit (`systemctl cat
cloudflared` ? the
40     `--token ?` value). You'll copy it to the VM in $6.
41     - **For the WASB path only:** the weights at `gs://khelsutra-rally-
corpus/models/wasb_badminton_best.pth.tar`.
42
43     You'll work in **two repo checkouts** on your machine: this one (`khelsutra`) for the
config templates,
44     and the **engine repo** (`badminton-highlight-indexer`) for the GPU-VM scripts. Commands
below say which.
45
46     ---
47
48     ## 1. Provision the L4 VM *(engine repo)*
49
50     ```bash
51     # in the badminton-highlight-indexer checkout
52     RALLY_VM_NAME=khelsutra-gpu RALLY_GPU=L4 bash deploy/gcp/create_gpu_vm.sh
53     ```
54
55     It sweeps zones (asia-south1 first ? Mumbai, near your users + co-located with the
bucket), uses a
56     driver-ready Deep-Learning-VM image (so `nvidia-smi` works immediately), and bakes in
the Ops Agent.
57     **Note the zone it prints** (e.g. `asia-south1-b`) ? you need it for SSH/teardown.
58
59     ```bash
60     ZONE=<the-zone-it-printed>

```

```

61     gcloud compute ssh khelsutra-gpu --zone=$ZONE    # confirm: nvidia-smi shows the L4
62     ```
63
64     ---
65
66     ## 2. Put the engine on the VM
67
68     The engine repo is private and the VM has no git creds, so push the source from your
machine
69     (byte-for-byte the same trick the droplet used):
70
71     ```bash
72     # in the badminton-highlight-indexer checkout
73     git archive --format=tar origin/master \
74         | gcloud compute ssh khelsutra-gpu --zone=$ZONE --command \
75         'sudo mkdir -p /opt/khelsutra/engine && sudo chown -R $USER /opt/khelsutra && tar -x
-C /opt/khelsutra/engine'
76     ```
77
78     Then on the VM (`gcloud compute ssh khelsutra-gpu --zone=$ZONE`):
79
80     ```bash
81     # ffmpeg + cv2 libs. The DLVM image is Ubuntu 22.04 ? libgl1 libglv2 libglx0 (on Ubuntu 24.04 it's
libgl1 libglv2 libglx0).
82     sudo apt-get update -qq && sudo apt-get install -y ffmpeg libgl1 libglv2 libglx0
83     sudo mkdir -p /srv/khelsutra/incoming && sudo chown -R "$USER" /srv/khelsutra
84     cd /opt/khelsutra/engine
85     python3 -m venv /opt/khelsutra/venv
86     /opt/khelsutra/venv/bin/pip install -q --upgrade pip
87     /opt/khelsutra/venv/bin/pip install -q -e '[auth,storage-gcs]'    # auth = the Cf-Access
JWT verify
88     ```
89
90     `backend.main` is torch-free, so this installs fast. (Torch only comes in with the WASB
path, $3.)
91
92     > **? Run everything as ONE user (don't reuse the droplet's `khelsutra` system user
here).** On the
93     > droplet the engine ran as a locked-down `User=khelsutra`; on this GPU VM the simplest,
mistake-proof
94     > setup is to keep the engine, the venv, **and the WASB conda env ($3) all owned by your
VM login user**
95     > (`whoami` after `gcloud compute ssh`, e.g. your Google id). A split owner is the #1
footgun here:
96     > the WASB env lands in `~/miniconda3` (your home), which a different service user can't
read, so a
97     > WASB run fails with a cryptic permission/`WASB_PYTHON not found`. $4 sets the systemd
`User=` to this
98     > same login user. (If you insist on the hardened `khelsutra` user, you must install the
venv **and**
99     > the conda env into a path it can read, e.g. under `/opt`, and `chown` them to
`khelsutra`.)
100
101     ---
102
103     ## 3. *(WASB path only)* Build the native GPU detection env
104
105     Skip this if you'll only run Gemini. For WASB-on-the-L4:
106
107     ```bash
108     # on the VM, in /opt/khelsutra/engine
109     bash deploy/digitalocean/gpu_setup.sh    # builds the `wasb` conda env
(~/.miniconda3/envs/wasb, torch 1.11+cu113, clones WASB-SBDT)
110     mkdir -p ~/models/WASB-SBDT/pretrained_weights
111     gcloud storage cp gs://khelsutra-rally-corpus/models/wasb_badminton_best.pth.tar \

```

```

112     ~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
113     ``
114
115     The native runner needs three env vars (set them in `engine.env`, §4):
`WASB_PYTHON`,
116     `WASB_REPO_DIR`, `WASB_WEIGHTS`. `WASB_WEIGHTS` alone is not enough (validated on an
L4, 2026-06-22).
117
118     ---
119
120     ## 4. Serving config + start
121
122     Reuse the droplet templates from this repo's [`deploy/droplet/`](droplet/) ? the AUD,
team domain,
123     and hosts are identical to the droplet (it's the same tunnel + Access app).
124
125     ``/srv/khelsutra/config.json`` (from `config.json.example`): keep
`security.edge_auth_enabled=true`,
126     `allowed_video_root=/srv/khelsutra/incoming`, your real `cf_team_domain` +
`cf_access_aud`. Set
127     `indexing.default_segmenter` per §5.
128
129     ``/etc/khelsutra/engine.env`` (from `engine.env.example`):
130     ```bash
131     GEMINI_API_KEY=<your rotated key>           # for the Gemini path
132     RALLY_APP_HOME=/srv/khelsutra
133     RALLY_ALLOWED_HOSTS=api.khelsutra.guru,localhost,127.0.0.1  # REQUIRED (Host-guard) ?
load-bearing
134     RALLY_BIND_HOST=127.0.0.1                  # cloudflared dials 127.0.0.1:8000
135     RALLY_CORS_ALLOW_ORIGINS=https://app.khelsutra.guru
136     # --- WASB path only (add these). Use YOUR login user's real home ? print it with: echo
$HOME ---
137     WASB_PYTHON=$HOME/miniconda3/envs/wasb/bin/python
138     WASB_REPO_DIR=$HOME/models/WASB-SBDT
139     WASB_WEIGHTS=$HOME/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
140     ``
141     > systemd does not expand `$HOME` ? paste the literal path (`echo $HOME` ? e.g.
`/home/yourid`).
142
143     ``/etc/systemd/system/khelsutra-engine.service`` ? start from
`deploy/droplet/khelsutra-engine.service`,
144     but make two edits for this VM:
145     - `User=` ? your VM login user (`whoami`), NOT `khelsutra` (see the §2 one-user
callout);
146     - `WorkingDirectory=/opt/khelsutra/engine` (matches §2).
147
148     Then:
149
150     ```bash
151     sudo systemctl daemon-reload && sudo systemctl enable --now khelsutra-engine
152     # local verification (still serving via the droplet at this point):
153     curl -s localhost:8000/api/health           # 200
154     curl -s localhost:8000/api/capabilities | python3 -m json.tool  # check segmenters +
gpu.nvidia_smi_present
155     ``
156
157     If it refuses to start, that's the boot-posture guard doing its job ? fix the
AUD/key/hosts and retry.
158
159     ---
160
161     ## 5. Switch detection: Gemini ? WASB
162
163     The server default is one config flip + restart. (API callers can also override per
request by sending

```

```

164     `"segmenter": "wasb_hybrid"` in `POST /api/process`.)
165
166     | | **Gemini** (proven, no GPU) | **WASB native** (uses the L4) |
167     |---|---|---|
168     | `config.json` ? `indexing.default_segmenter` | `"gemini"` | `"wasb_hybrid"` |
169     | `config.json` ? `indexing` also | ? | `"detector_impl": "native", "skip_ai_handoff":
true` |
170     | `engine.env` needs | `GEMINI_API_KEY` | the 3 `WASB_*` vars ($3/$4) |
171     | Prereq | a Gemini key | the WASB env + weights ($3) |
172     | GPU used? | no (cloud API) | **yes** |
173     | Per-match cost | Gemini API tokens | none (pure-engine) |
174
175     Apply: edit `config.json`, then `sudo systemctl restart khelsutra-engine`. Confirm with
176     `curl -s localhost:8000/api/capabilities` (`default_segmenter`) and a real run.
177
178     > Start on **Gemini** (it's exactly what's live now), prove the VM serves end-to-end,
cut the tunnel
179     > over ($6), then flip to **WASB** once you're happy ? so you change one variable at a
time.
180
181     ---
182
183     ## 6. Cut the tunnel over to the VM
184
185     The tunnel `khelsutra-studio` is **token-managed**: its ingress (`api.khelsutra.guru` ?
`localhost:8000`)
186     lives in the Cloudflare dashboard and follows whichever machine runs the connector. So:
187
188     ```bash
189     # 1. get the token from the droplet (it's your secret; don't paste it anywhere public)
190     ssh root@<droplet-ip> 'systemctl cat cloudflared' | grep -o -- '---token
[A-Za-z0-9._=-]*'
191
192     # 2. on the VM: install cloudflared + run the SAME token
193     gcloud compute ssh khelsutra-gpu --zone=$ZONE
194     curl -fsSL https://pkg.cloudflare.com/cloudflared-stable-linux-amd64.deb -o
/tmp/cf.deb && sudo dpkg -i /tmp/cf.deb
195     sudo cloudflared service install <THE_TOKEN>          # installs + starts the connector as
a service
196     sudo systemctl status cloudflared                    # "Registered tunnel connection" ?
the VM is now a replica
197     ```
198
199     At this point **both** the droplet and the VM serve the tunnel (Cloudflare HA ? either
can answer; both
200     run the same engine, so that's fine). Verify the VM is healthy through the public
hostname:
201
202     ```bash
203     curl -sI https://api.khelsutra.guru/api/health        # 200 (or the 302?Access login if
unauthenticated ? that's correct)
204     ```
205
206     Then **stop the droplet's connector** so all traffic lands on the VM:
207
208     ```bash
209     ssh root@<droplet-ip> 'systemctl stop cloudflared && systemctl disable cloudflared'
210     ```
211
212     DNS, the CF-Access app, and the dashboard ingress are unchanged ? only the connector
host moved.
213
214     ---
215
216     ## 7. Verify the live app is on the VM

```

```

217
218     - `https://app.khelsutra.guru` ? log in (email OTP) ? upload a match ? process ? pick
rallies ? **Build**
219     ? download. (Compile is async ? it polls; a few minutes is normal.)
220     - The engine's logs are now on the VM: `gcloud compute ssh khelsutra-gpu --zone=$ZONE
--command 'journalctl -u khelsutra-engine -n 50 --no-pager'`.
221     - Per-job proof of which detector ran: `GET /api/jobs/<id>` ? `result.compute.path` /
the segmenter in the message.
222
223     ---
224
225     ## 8. Rollback (if anything's off)
226
227     The droplet is untouched (engine still installed, just its cloudflared stopped):
228
229     ```bash
230     ssh root@<droplet-ip> 'systemctl enable --now cloudflared'    # droplet serves again
231     gcloud compute ssh khelsutra-gpu --zone=$ZONE --command 'sudo systemctl stop
cloudflared'    # VM stops serving
232     ```
233
234     (Run the droplet start **before** stopping the VM for zero-gap rollback.)
235
236     ---
237
238     ## 9. Cost control / teardown
239
240     - **Stop billing entirely:** `RALLY_GCP_ZONE=$ZONE bash deploy/gcp/teardown.sh delete`
(engine repo) ?
241     **delete**, never *stop* (a stopped VM still bills its ~$15?35/mo disk).
242     - Leaving it up is ~$500?600/mo. If load is intermittent, consider the scale-to-zero
Cloud Run GPU
243     *service* (backlogged) instead.
244     - Observability: the VM has the Ops Agent ? GPU/CPU/mem graphs are in Cloud Monitoring.
245
246     ---
247
248     ### Quick reference
249
250     | Thing | Where |
251     |---|---|
252     | Provision / teardown the L4 VM | engine repo
`deploy/gcp/{create_gpu_vm.sh,teardown.sh}` |
253     | WASB conda env builder | engine repo `deploy/digitalocean/gpu_setup.sh` |
254     | Serving config templates | this repo
`deploy/droplet/{config.json.example,engine.env.example,khelsutra-engine.service}` |
255     | Tunnel ingress (managed in CF dashboard) | Zero Trust ? Networks ? Tunnels ?
`khelsutra-studio` |
256     | The original droplet deploy | [DEPLOY_ALPHA.md](DEPLOY_ALPHA.md) |
257

```

15. user (2026-07-02T04:50:37.071Z)

```

1     # Droplet quickstart ? KhelSutra alpha (the `api.` engine side)
2
3     The **filled-in, command-dense** version of `../DEPLOY_ALPHA.md` for the Linux CPU
droplet
4     (`168.144.126.176`, Gemini, **no GPU**). Values are pre-baked for **`app.khelsutra.guru`
/
5     `api.khelsutra.guru`**, team domain **`https://khelsutra.cloudflareaccess.com`**,
allowlist
6     **`avi.dullu@gmail.com`**, jail **`/srv/khelsutra/incoming`**. Read `../DEPLOY_ALPHA.md`
for the *why*;
7     this is the *what to type*. Files here:
8

```

```

9      | File | Goes to | Purpose |
10     |-----|-----|-----|
11     | `config.json.example` | `/srv/khelsutra/config.json` | the `security` block (edge-auth
+ jail + CF team/AUD) |
12     | `engine.env.example` | `/etc/khelsutra/engine.env` | secrets + the two `RALLY_`
exposure vars + app-home |
13     | `khelsutra-engine.service` | `/etc/systemd/system/` | keeps the engine up (binds
`127.0.0.1:8000`) |
14     | `../cloudflared.config.example.yml` | `/etc/cloudflared/config.yml` | the tunnel
ingress (`api.` ? `127.0.0.1:8000`) |
15
16     > Cost reality: Cloudflare Pages + Tunnel + Access are free (Access ? 50 users).
No $25/Pro
17     > plan. When you enable Zero Trust you may be asked to add a card and "subscribe" ?
pick the Free
18     > plan; you won't be charged under 50 users. Standing cost ? $0 + Gemini per run.
19
20     ---
21
22     ## A. Engine on the box (one-time)
23
24     ```bash
25     # 1. user + jail dirs (upload_dir MUST sit under allowed_video_root)
26     sudo useradd --system --create-home --home-dir /opt/khelsutra khelsutra || true
27     sudo mkdir -p /srv/khelsutra/incoming/uploads
28     sudo chown -R khelsutra:khelsutra /srv/khelsutra /opt/khelsutra
29
30     # 2. engine code + venv + the AUTH extra (REQUIRED with edge auth on, else every authed
request 401s)
31     sudo -u khelsutra git clone <ENGINE_REPO_URL> /opt/khelsutra/badminton-highlight-indexer
32     sudo -u khelsutra python3 -m venv /opt/khelsutra/venv
33     sudo -u khelsutra /opt/khelsutra/venv/bin/pip install -e "/opt/khelsutra/badminton-
highlight-indexer[auth]"
34     # (ffmpeg is required by the engine: `sudo apt-get install -y ffmpeg`)
35
36     # 3. env file (fill GEMINI_API_KEY) ? keep it 600
37     sudo mkdir -p /etc/khelsutra
38     sudo cp deploy/droplet/engine.env.example /etc/khelsutra/engine.env
39     sudo nano /etc/khelsutra/engine.env # paste the rotated GEMINI_API_KEY
40     sudo chmod 600 /etc/khelsutra/engine.env
41
42     # 4. config.json at the app-home path the server actually reads
($RALLY_APP_HOME/config.json)
43     sudo -u khelsutra cp deploy/droplet/config.json.example /srv/khelsutra/config.json
44     # cf_access_aud is still a placeholder ? you fill it in step C after creating the
Access app.
45
46     # 5. service
47     sudo cp deploy/droplet/khelsutra-engine.service /etc/systemd/system/
48     # ^ edit WorkingDirectory + ExecStart paths if you didn't use /opt/khelsutra/...
49     sudo systemctl daemon-reload && sudo systemctl enable --now khelsutra-engine
50     journalctl -u khelsutra-engine -e # expect a clean boot; a WARN about
cf_access_aud is fine until step C
51     ```
52
53     The boot fails fast (`exit 1`, `[STARTUP] refusing to start`) if
`edge_auth_enabled=true` but the
54     jail or `cf_team_domain`/`cf_access_aud` is unset ? that's your signal the config is
read & wrong, not ignored.
55     If it boots silently claiming auth-on, the engine read a different `config.json`
(check `RALLY_APP_HOME`).
56
57     ## B. The tunnel (`api.khelsutra.guru` ? the box)
58
59     ```bash

```

```

60     # install cloudflared (Cloudflare apt repo), then:
61     cloudflared tunnel login                                # browser OAuth to the
khelsutra.guru zone
62     cloudflared tunnel create khelsutra-studio              # note the <TUNNEL_UUID> it
prints
63     sudo mkdir -p /etc/cloudflared
64     sudo cp deploy/cloudflared.config.example.yml /etc/cloudflared/config.yml
65     sudo nano /etc/cloudflared/config.yml                    # put <TUNNEL_UUID> in BOTH
lines (tunnel: + credentials-file:)
66     sudo cp ~/.cloudflared/<TUNNEL_UUID>.json /etc/cloudflared/ # move the creds where
config.yml points
67     cloudflared tunnel route dns khelsutra-studio api.khelsutra.guru # creates the api.
CNAME
68     sudo cloudflared service install                        # run it as a service
69     ```
70
71     ## C. Cloudflare dashboard (?10 min, click-ops)
72
73     1. **Pages** ? Workers & Pages ? Create ? Pages ? Connect to Git ? `khelsutra` repo.
74         - Root dir `web` · Build `npm run build` · Output `web/dist`
75         - Env var **`VITE_API_BASE = https://api.khelsutra.guru/api`**
76         - Custom domain ? add **`app.khelsutra.guru`**.
77     2. **Zero Trust ? Access ? Applications ? Add ? Self-hosted**
78         - Application domains: **both** `app.khelsutra.guru` **and** `api.khelsutra.guru`
(one app ? one session).
79         - Policy: *Allow* ? *Emails* ? **`avi.dullu@gmail.com`** (one-time-PIN; add tester
emails later).
80         - **Settings ? enable "Bypass OPTIONS requests to CORS preflight" ? load-bearing;
without it every cross-origin call dies at the preflight.
81         - Copy the application **AUD** ? paste into `/srv/khelsutra/config.json`
`security.cf_access_aud` ? `sudo systemctl restart khelsutra-engine`.
82
83     ## D. Verify (must hold before sharing)
84
85     ```bash
86     # 1. from OUTSIDE the box ? refused (engine is 127.0.0.1-bound; you also confirmed no
inbound :8000)
87     curl --max-time 5 http://168.144.126.176:8000/api/health      # -> refused / timeout
88
89     # 2. the gate is in front (no CF session) ? a redirect to login, NOT a 200
90     curl -i https://api.khelsutra.guru/api/health                # -> 302 to the Access
login
91
92     # 3. the Host guard passes for the tunnel host (this is what RALLY_ALLOWED_HOSTS fixes;
would 400 without it)
93     #     after signing in via the browser, app.khelsutra.guru loads and /api/health returns
200.
94     ```
95
96     **The shareable CUJ:** open `app.khelsutra.guru` ? sign in (email OTP) ? **upload a
video** (lands in the
97     jail) ? Process ? pick rallies ? Build ? download the reel. *(Pasting a `gs://` URI is
rejected by the
98     jail on the exposed box ? upload is the alpha path; see `../DEPLOY_ALPHA.md` §7.)*
99
100    > **? Live-verified on the droplet (2026-06-22).** Engine installed + booted on
`168.144.126.176`; full
101    > suite **1313 passed** + ruff/mypy clean on the box; boot-posture fail-fast, the Host-
guard `400?200`,
102    > the path-jail (`gs://`/etc/passwd`/`..` ? 400), the T4 upload?jail, and the edge-
auth `401`-gate all
103    > exercised live. Tunnel + Access + Pages are up; the full **DNS ? Cloudflare ? Access ?
tunnel ? engine**
104    > chain is confirmed (`/api/health` = 200 through an Access session; unauthenticated ?
302 to the login).

```

```

105     > Two things still need a human: the rotated **Gemini key** (`etc/khelsutra/engine.env`
? restart) and a
106     > **browser one-time-PIN login** to exercise the upload?reel CUJ.
107     >
108     > **? Two gotchas the live run surfaced:**
109     > - **`indexing.default_segmenter` MUST be `gemini` on a CPU box** (set in
`config.json.example`). The SPA
110     > sends no segmenter, so the engine uses its default ? the shipped default
`yolo_hybrid` needs torch/GPU
111     > and every job fails `No module named 'torch'` without this.
112     > - **Without a valid `GEMINI_API_KEY`, a job finishes `done` with 0 rallies** ? the
missing-key error lands
113     > only in the engine log, not the job status. Set the key before sharing.
114

```

16. user (2026-07-02T04:50:44.383Z)

```

35     | D6 | Compute placement is a **setting**, rendered from `GET /api/capabilities` |
`[design]`; 2026-06-20 | UI offers ONLY what the box can honestly do |
36     | D7 | **Ingest is a source menu** (local upload · bucket URI · Drive) normalizing to
one video object | `[verified backend/storage/ref.py parse_uri]`; 2026-06-20 | Drive is a
**stub** today ? scoped honestly ($4.2) |
37     | D8 | GPU host is **provider-agnostic** (DO droplet / GCE VM / Cloud Run+GPU); storage
stays the bucket | Owner 2026-06-20 | compute and storage are separate planes; no provider lock-
in |
38     | D9 | **Observability is first-class** ? structured logs on every new path + a DoD item
| Owner 2026-06-20 | see $5; verified during CUJ replay |
39     | D10 | Delivery = a **self-hosted webserver + UI the user spins up** (NOT a desktop
app); the user points at **local storage + GPU/CPU OR cloud locations** and the UI drives the
tool | Owner 2026-06-20 | **supersedes `DESKTOP_APP_PLAN.md`**; one artifact (this appliance)
runs on the owner box, a droplet, or a GPU host ? same UI, compute/storage are settings |
40
41     ## 3. Foundation ? what the engine gives us today `[verified this session unless
noted]`
42     - **FastAPI loop:** `POST /api/process` (202 + `job_id`) ? poll `GET /api/jobs/{id}` ?
`POST /api/query` ? `POST /api/compile` ? `GET /api/highlights/{video_id}/{filename}`
(FileResponse) `[main.py]`. **Chunked upload exists** ? `POST /api/upload/init` · `PUT
/api/upload/{id}/part/{i}` · `POST /api/upload/{id}/complete` · `DELETE /api/upload/{id}`
`[verified uploads.py:64,89,123,147]` ? but the bundled `app.js` **never calls them** (it only
hits `/api/process` + `/api/compile` `[verified app.js:88,373]`). So browser upload is **client-
wiring only**, no engine change.
43     - **In-process worker:** one module-global `ThreadPoolExecutor(max_workers=1)`
`[verified job_worker.py:37]` ? strictly **one job at a time**; the DB `jobs` table is the
clock; `fail_stale_jobs` sweeps queued/running on boot. Gemini chunk concurrency is a
**separate** knob `indexing.ai_max_workers` (**default 5** `[verified models.py:183]`; set to 1
to serialize) ? it is **not** already 1.
44     - **Storage:** `storage.backend` default = `"local"` `[verified models.py:298]`;
`LocalStorage.fetch_to_local` is **identity** `[verified local.py:33]`, `put` no-ops on
`src==dst` ? so on a single box the worker's pull?pipeline?push **collapses to read-local/write-
local**. URI scheme `parse_uri` `[verified ref.py]`: bare/`local://` ? local; `s3://`
(AWS/R2/**DO Spaces**/MinIO, by endpoint) ; `gcs://`/`gs://` ; `gdrive://` ? **REAL (shipped
2026-07-01): a mounted Google-Drive-for-Desktop alias* (`storage/gdrive.py`) + a Drive-API
backend for headless boxes (`storage/gdrive_api.py`). Credentials never in config ? boto3
default chain / `~/aws`, Drive via ambient OAuth `[verified s3.py, gdrive.py]`.
45     - **Compute seam:** `indexing.detector_impl` ? `{auto`=WSL, `native`=Linux-native,
`stub`=CPU mock}; default `"auto"` `[verified models.py:129]`, resolved in
`trajectory_hybrid._resolve_runner` `[verified :76-85]`. **Native WASB `native_wasb_runner.py`
is now on `origin/master` (no longer branch-only; `tests/test_native_wasb_runner.py` is on
master too) `[verified git ls-tree origin/master]`; **`fusion_hybrid._build_native_runner`
returns `NativeWasbRunner` **`[verified fusion_hybrid.py:86, origin/master]` while
**`trajectory_hybrid._build_native_runner` still raises `NotImplementedError` **`[verified
trajectory_hybrid.py:61, origin/master]`. Acceptance gate = **owner byte-parity** (WSL CSV ==
native CSV) on a real GPU box ? not exercisable on the droplet or in the offline suite. Treat
native as **not production-verified**.
46     - **Worker CLI:** `python -m backend.worker {infer|train}` pulls a `--video-uri` ?

```

```

pipeline ? pushes `outputs/<video_id>/{intervals.csv,report.json}` `[verified
worker/___main__.py:88,149]`. The validated GPU-free/key-free combo: `detector_impl=stub` +
`--set indexing.skip_ai_handoff=true` + `storage.s3`.
47 - **Gating:** cloudflared Tunnel (outbound-only, no inbound ports) + Cloudflare Access
email-OTP; backend trusts `Cf-Access-Authenticated-User-Email` `[HOSTING_PLAN.md]`. `GET
/api/health` **now EXISTS** (engine M1, [#268]) ? `{status, sport, default_segmenter}`
`[verified main.py:127, corrected 2026-06-21 ? was "does not exist"]`.
48 - **Droplet reality:** `168.144.126.176`, **CPU-only, NO GPU**, region `blr1`; co-hosts
the marketing site (`/srv/khelsutra-site`, node `:8787`) **and** the engine (`~/rally`); ran the
full train+infer e2e against DO Spaces with the **GPU mocked** `[SETUP_NEW_MACHINE.md,
DECOUPLED_COMPUTE/HANDOFF.md]`. Shared testbed: MinIO `9000/9001` + GCS emulator `9023` also run
there.
49 - **Shipped since this doc's first draft (all on `origin/master`):** `GET
/api/capabilities` (now also carries a **`storage` block** ? usable media local/s3/gcs/gdrive +
free space, P3) . `api/health` . `api/version` . **`GET /api/videos`** (the "my videos" list,
P6) . **`GET /api/source/{id}`** (a frame-identical annotation proxy, D4) . a per-video **cost
ledger** (`/api/jobs/{id}/cost`, `/api/videos/{id}/cost`) . **`POST /api/annotations`** (P8) .
**`POST /api/assets`** (store-into-medium, P4) . a **free-space guard** (HTTP 507 on
upload/fetch; WSL frame-stage now BLOCKS, P2).
50 - **Still absent (net-new):** `GET /api/reels/{video_id}` (per-video reel list; the
global `api/reels` exists but the SPA doesn't consume it), any **Dockerfile** `[verified ? grep
returns none]`.
51
52 ## 4. Design / architecture
53
54 ### 4.1 Topology & request path
55 ONE box: **edge gate** (cloudflared+Access or Caddy basicauth) ? **reverse proxy /
static host** serving the `web/` SPA bundle and proxying `api/*` ? engine **FastAPI
`127.0.0.1:8000`** ? **LocalStorage** ? **detector seam** (Gemini/stub today; native WASB on a
GPU host). The firewall blocks inbound `8000` + the test ports `9000/9001/9023`. The marketing
site stays a **separate** systemd unit on a **separate** subdomain.
56
57 ```
58 PUBLIC INTERNET
59 ? (1) EDGE GATE ? the ONLY ingress; auth lives HERE, not in the app
60 ? cloudflared Tunnel + Cloudflare Access (no inbound ports) OR Caddy :443 +
basicauth
61 ?
62 ?????????????????????????????????????????????????????????????????????????????????
63 ? ONE BOX firewall BLOCKS inbound 8000 + 9000/9001/9023 ?
64 ? (2) reverse proxy / static host (Caddy/cloudflared) ?

```

17. user (2026-07-02T04:50:45.580Z)

```

160 | T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key
(P1) | ? | owner | ? | ? |
161 | T3 | L3 live posture test (raw `:8000`/:900x` refused from outside) | T2 | no | ? | ?
|
162 | T4 | S1 ingest: **local upload wired ?
([#66](https://github.com/avidullu/khelsutra/pull/66))** ? file picker ? chunked `api/upload`**
? server path ? `api/process` ? existing poll?load?compile?download; segmenter picker still
pending | T1 | no | ? | [#66](https://github.com/avidullu/khelsutra/pull/66) |
163 | T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
164 | T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
165 | T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner ? **engine half
shipped on `origin/master`** (`api/capabilities` main.py:340, `api/health` :307); only the SPA
banner remains | T1 | coord | ? | engine (on master) |
166 | T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) |
? | coord | ? | ? |
167 | T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no |
? | ? |
168 | T10 | Observability pass: correlation id end-to-end + structured logs + metrics (S5) |
T4 | no | ? | ? |
169 | T11 | Retention sweep for `output/`+`uploads/` ? **sweep tool shipped** ([engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299)): dry-run-first; purges

```

stale uploads >7d + orphan `.partial-\*` + regenerable frames/staged/proxies; **reels/DB/CSVs durable** ; `--json` plan ready for a scheduler. Automated scheduling ? [engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301). | ? | owner ? (uploads 7d, keep reels ? 2026-06-21) | ? | [engine #299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) |

170 | T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? engine accepts URIs (M2/[#280]) ** SPA ingest screen shipped (B-P2, [#56])**; in-UI `gs://`?reel CUJ proven live e2e (`DRY\_RUN\_CUJ.md` §9) | T4 | ? | ? | engine #280 + SPA #56 |

171 | T13 | **GPU host stand-up** (provider-agnostic) + wire `detector\_impl=native`; owner byte-parity (P4) | branch | ? owner | ? | ? |

172 | T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) | T8,T12 | ? owner | ? | ? |

173

174 **Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the start** ? owner provisions the host + token; the engine session lands native WASB; the UI MVP does **not** block on it (ships on Gemini).

175

176 ** ? Alpha deploy path chosen = SPLIT, not the single-origin shell (owner 2026-06-21).** The alpha ships the **split** topology (SPA on Cloudflare Pages + engine on an `api.` cloudflared tunnel + CF Access over both ? the engine `HOSTING\_PLAN.md` §2), **not** the single-origin Caddy shell of **T2** (kept as the documented alternative ? §0/D1). The **deploy kit + runbook** ([`deploy/DEPLOY\_ALPHA.md`](../deploy/DEPLOY_ALPHA.md) + [`deploy/cloudflared.config.example.yml`](../deploy/cloudflared.config.example.yml)) is shipped (? templates+runbook). **The alpha stand-up is now LIVE (2026-06-22):** the engine booted on the droplet (`168.144.126.176`), `app.khelsutra.guru` is up, and the full **DNS ? Cloudflare ? Access ? tunnel ? engine** chain plus the **T3** raw-`:8000`-refused posture test (boot-posture fail-fast, Host-guard `400?200`, path-jail, T4 upload?jail, edge-auth `401`-gate) were all **exercised live** [`deploy/droplet/README.md:100-106`]. Only the **Gemini-key rotation** + a **browser OTP login** remain. Enabled by: engine exposure-safety boot posture (engine #294) + SPA `VITE\_API\_BASE` ([#64]).

177

178 ## 10. Open questions ? owner / external

179 **Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention defaults** ? **window DECIDED** (uploads **7d**, reels durable; [engine #299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) tool + scheduling [#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)); **still open:** auto-delete originals after compile vs keep proxies+intervals only (privacy/consent). (3) **per-video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4) **GPU host choice** ? DO GPU droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker worth building for Cloud Run)? (5) **P5 bridge ownership** ? who builds the orchestration endpoint + how it coordinates with the active worker sessions.

180 **Nice-to-knows:** console home (web/ vs engine static); a review-playback streaming endpoint; per-user scoping timing; containerization vs runbook.

181

182 ## 11. Definition of done

183 - [~] CUJ-1 works end to end on the droplet (Gemini **and** stub) behind the gate, via the ingest menu. *(2026-06-22: chain LIVE end-to-end ? DNS?Access?tunnel?engine, T4 upload?jail exercised; the full upload?reel pass still needs the Gemini key + a browser OTP login [`deploy/droplet/README.md:100-106`]).*

184 - [x] Raw `:8000`/`:9000` unreachable from outside (**tested** live 2026-06-22, not assumed) [`deploy/droplet/README.md:100-106`].

185 - [] Local upload wired; bucket-URI ingest at least designed/tracked; Drive scoped honestly (not faked).

186 - [] RallyPicker embedded with honesty rules; `min\_shot\_count` surfaced.

187 - [] `/api/capabilities` + `/api/health` + banner truthful; reload-survival via `/api/videos` + `/api/reels`.

188 - [] **Observability:** correlation id threaded end-to-end; structured logs at every new path; loud failures; logs inspected during CUJ replay.

189 - [] Retention sweep in place (**tool shipped** [engine #299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299); **auto-schedule** [engine #301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)); `GEMINI\_API\_KEY` rotated; `allowed\_video\_root` set.

190 - [] CI green (L2 contract + L4 regression); coverage ? floor; §9 tracker all ??-resolved.

```

191
192     ## 12. References
193     **Engine code `[verified this session]`:**
194     `backend/main.py:12-17,42,52,307-328,331-340,342-396,638`;
195     `backend/api/uploads.py:64,89,123,147`; `backend/api/job_worker.py:37`;
196     `backend/api/static/app.js:88,344,373`;
197     `backend/storage/{base.py,local.py:33,ref.py,__init__.py:43}`;
198     `backend/config/models.py:129,183,298`;
199     `backend/pipeline/segmenters/{trajectory_hybrid.py:57-63,76-85, fusion_hybrid.py:76-86,
200     gemini.py:470-485, motion.py:191-198}`; `backend/pipeline/detectors/{native_wasb_runner.py,
201     stub_runner.py}`; `backend/worker/__main__.py:88,98,149`; `config.json` (`default_segmenter:
202     gemini`). **Engine docs:** `HOSTING_PLAN.md:42,102-108`; `VIDEO_LOCALITY_MODEL.md`;
203     `DESKTOP_APP_PLAN.md`; `SETUP_NEW_MACHINE.md`; `PLATFORM_ARCHITECTURE.md`;
204     `DECOUPLED_COMPUTE/{README,02,04,07,HANDOFF}.md`; `UI_PROPOSAL.md:41-66`;
205     `PROJECT_DOC_TEMPLATE.md`. **This repo:** `PER_RALLY_SELECTION.md`, `REQUEST_ACCESS_FORM.md`,
206     `PORTABILITY.md`, `web/README.md`.
207
208     ---
209     ### Changelog
210     - 2026-06-22 ? Freshness pass: `DRAFT`?`IN PROGRESS`; re-anchored engine facts against
211     `origin/master` ? `/api/capabilities` + `/api/health` now exist (T7 engine half ?);
212     `native_wasb_runner.py` is on master (no longer branch-only); CORS `` is now env-lockable + the
213     engine ships a Host-guard / optional Cf-Access edge-auth / boot-posture (so "zero app-layer
214     auth" is no longer literal); alpha deploy went LIVE 2026-06-22 (`app.khelsutra.guru`), ticking
215     the raw-`:8000`-refused DoD box.
216     - 2026-06-20 ? Initial `DRAFT`. Consolidated three single-box-appliance designs
217     (FastAPI-native / unified-origin / decoupled-profile) into the reverse-proxied unified-origin
218     spine + compute-as-a-setting framing, via a multi-agent design pass; every engine-fact anchor
219     re-verified against source before writing (catching the native-WASB partial-wiring +
220     `ai_max_workers` default nuances). Folded in the owner's ingest-menu (Drive/bucket/local),
221     multi-provider GPU, and first-class Observability directives.
222
223

```

18. user (2026-07-02T04:50:52.915Z)

```

30     1. **Engine host + deps:** run the engine FastAPI (`python -m backend.main --server`,
31     binds `127.0.0.1:8000`) on a box with `ffmpeg` + OpenCV. Gemini/motion paths are **CPU-only, no
32     WSL** ? local Windows works; the CPU droplet works too. (GPU/WSL only needed for native WASB.)
33     2. **Get the bucket video local + into the API DB:**
34     `gsutil cp gs://<bucket>/<match>.mp4 ./incoming/` (or the worker fetch), then `POST
35     /api/process {"video_path":"<abs local path>","segmenter":"gemini"}` ? poll `GET
36     /api/jobs/{job_id}` until `result.video_id`. *(`gemini` needs `GEMINI_API_KEY`; `motion` is key-
37     free but hides fabricated metadata.)*
38     3. **Open the SPA against the engine:** `npm run dev` (Vite proxies `/api` ? `:8000`,
39     same-origin) ? `http://localhost:5173/?video=<video_id>` ? rallies auto-load.
40     4. **Select ? Build ? download:** pick rallies ? Build ? `<video>` preview ? click the
41     download link ? MP4 saved local.
42     5. **Repeat in Hindi:** click `????????` (or `?lang=hi`) ? redo select?compile?download
43     ? confirm the whole flow works with the UI in Hindi.
44
45     **Small unblocked SPA polish that makes Path A more "via the UI" (independent of A/B,
46     useful in both): a ***Load a video by ID*** input (paste the `video_id` ? load, instead of
47     editing the URL) + consume the engine's `download_url` (instead of hand-building it). One tight
48     PR.
49
50     ## 4. Open decisions ? OWNER (these gate the dry run)
51     1. **Scope:** Path A (cheapest dry run now) or Path B (the in-UI bucket ingest ? **SPA-
52     only**); the engine endpoint already accepts `gs://`, see §1)? *Recommend A first* (proves the
53     multilingual workflow fast), then B as the appliance build. ? ? **RESOLVED: A first (done, §7),
54     then B (B-P2, the SPA ingest/progress screen).**
55     2. **Compute / detector:** **Gemini** (real rallies; needs your `GEMINI_API_KEY`;
56     uploads downsized chunks to Google = paid + cloud) . **motion** (key-free, runs anywhere, but
57     lower-quality boundaries + fabricated server/shot metadata stays hidden) . **GPU WASB** (best
58     quality, but needs a GPU host + is owner-parity-gated, not production-verified). *Recommend
59     Gemini for "real-ish, cheapest" ? pending your key + OK (paid + uploads footage).*

```

```

42      3. **Engine host:** local Windows (simplest, Gemini/motion) · the CPU droplet
`168.144.126.176` (shared testbed ? careful) · a GPU VM (only for WASB).
43      4. **Which bucket video:** e.g. `gs://khelsutra-rally-
corpus/videos/mahadevpura_smoke_180s.mp4` (already staged, known-good) or another clean ?720p
clip (must pass the engine validators).
44
45      ## 5. Honest caveats
46      - **Gemini = paid + uploads footage to Google** (an external/paid action ? needs your
explicit OK + a clean clip). **Rotate the chat-shared `GEMINI_API_KEY`.
47      - **motion** fabricates `server`/`receiver`/`shot_count` (the SPA correctly hides them)
? boundaries are real but metadata isn't.
48      - **GPU/native WASB** is partially wired + owner-byte-parity-gated ? **not production-
verified**.
49      - Path A's "point at the bucket" is a **CLI pre-step**, not in-UI ? full in-UI ingest is
Path B.
50      - Edge auth: none in the engine (localhost dry-run is fine; required before any non-
localhost exposure).
51
52      ## 6. Definition of done (the dry run)
53      - [x] A bucket video ? (engine processes) ? the **SPA loads its rallies** ? select ?
compile ? **reel downloaded to the local machine**, all driven from the UI (Path A allows a one-
line CLI pre-process). ? ? **Path A done (local, 2026-06-21)**, see §7.
54      - [x] The **same flow works with the UI in Hindi** ? multilingual workflow confirmed. ?
? (Kannada also works via the same switcher.)
55      - [x] *(Path B = B-P2, SPA-only)* the bucket video is pointed-at **from the UI** (no CLI
pre-step). ? ? **DONE (live e2e, 2026-06-21), see §9.** In the SPA: entered a `gs://` URI ?
`POST /api/process` ? polled `/api/jobs` ? **12 rallies loaded** ? Build ? **reel downloaded
(1080p, 76s)**, all from the UI. Surfaced + fixed an engine compile bug along the way (engine
[#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)). **Screencast:**
`gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.
56      - [ ] *(option 3)* the same CUJ confirmed on the **CPU droplet** (Linux deploy target) ?
droplet reachable but currently **un-provisioned** (no engine checkout; ~3.8GB RAM) ? needs a
from-scratch engine install or a proper deploy of the fixed engine.
57
58      ## 7. ? Run result ? Path A, local (2026-06-21)
59      Ran the real CUJ end to end on the local Windows box, isolated engine off
`origin/master` (fresh DB/output worktree), **Gemini** detector:

```

19. user (2026-07-02T04:52:38.097Z)

```

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-
docrot\NEXT_STEPS.md:13:http://168.144.126.176:8787** (systemd `khelsutra-site`). Shipped across
~90 merged PRs (the form,
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-
docrot\deploy\DEPLOY_ALPHA.md:35:- A **box** to run the engine: the CPU droplet
`168.144.126.176` (Gemini-only ? no GPU; fine for alpha) **or** your Windows+WSL GPU box (real
WASB). `ffmpeg` + Python 3.8+ installed; the `badminton-highlight-indexer` engine checked out.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-
docrot\docs\DRY_RUN_CUJ.md:42:3. **Engine host:** local Windows (simplest, Gemini/motion) · the
CPU droplet `168.144.126.176` (shared testbed ? careful) · a GPU VM (only for WASB).
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-
docrot\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:48:- **Droplet reality:** `168.144.126.176`, **CPU-
only, NO GPU**, region `blr1`; co-hosts the marketing site (`/srv/khelsutra-site`, node `:8787`)
**and** the engine (`~/rally`); ran the full train+infer e2e against D0 Spaces with the **GPU
mocked** `[SETUP_NEW_MACHINE.md, DECOUPLED_COMPUTE/HANDOFF.md]`. Shared testbed: MinIO
`9000/9001` + GCS emulator `9023` also run there.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-
docrot\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:176:[Omitted long matching line]
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-

```

```
docrot\deploy\droplet\README.md:4:(`168.144.126.176`, Gemini, **no GPU**). Values are pre-baked
for **`app.khelsutra.guru` /
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-
docrot\deploy\droplet\README.md:87:curl --max-time 5 http://168.144.126.176:8000/api/health
# -> refused / timeout
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-
docrot\deploy\droplet\README.md:100:> *** Live-verified on the droplet (2026-06-22).** Engine
installed + booted on `168.144.126.176`; full
```

20. user (2026-07-02T04:52:38.990Z)

```
origin https://github.com/avidullu/khelsutra.git (fetch)
origin https://github.com/avidullu/khelsutra.git (push)
github.com
  ? Logged in to github.com account avidullu (keyring)
  - Active account: true
  - Git operations protocol: https
  - Token: gho_*****
```

21. user (2026-07-02T04:53:04.414Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/deploy/DEPLOY_ALPHA.md has been updated successfully. (file state is current in your context ? no need to Read it back)

22. user (2026-07-02T04:53:15.164Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/deploy/DEPLOY_ALPHA.md has been updated successfully. (file state is current in your context ? no need to Read it back)

23. user (2026-07-02T04:53:20.441Z)

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24. user (2026-07-02T04:53:24.490Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/deploy/DEPLOY_ALPHA.md has been updated successfully. (file state is current in your context ? no need to Read it back)

25. user (2026-07-02T04:53:31.680Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/deploy/DEPLOY_ALPHA.md has been updated successfully. (file state is current in your context ? no need to Read it back)

26. user (2026-07-02T04:53:35.874Z)

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27. user (2026-07-02T04:53:40.585Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/deploy/DEPLOY_ALPHA.md has been updated successfully. (file state is current in your context ? no need to Read it back)

28. user (2026-07-02T04:53:57.275Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

29. user (2026-07-02T04:54:03.796Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

30. user (2026-07-02T04:54:21.799Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

31. user (2026-07-02T04:54:41.532Z)

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32. user (2026-07-02T04:54:46.095Z)

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33. user (2026-07-02T04:54:53.174Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/deploy/GPU_VM_SERVING.md has been updated successfully. (file state is current in your context ? no need to Read it back)

34. user (2026-07-02T04:55:01.633Z)

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35. user (2026-07-02T04:55:14.336Z)

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36. user (2026-07-02T04:55:20.228Z)

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37. user (2026-07-02T04:55:29.448Z)

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indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/deploy/GPU_VM_SERVING.md has been updated successfully. (file state is current in your context ? no need to Read it back)

38. user (2026-07-02T04:55:34.355Z)

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39. user (2026-07-02T04:55:46.590Z)

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40. user (2026-07-02T04:56:05.186Z)

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41. assistant (2026-07-02T04:56:10.832Z)

Now update the two remaining line-number citations to the droplet README (they'd drift with my banner) and add a changelog entry.

42. user (2026-07-02T04:56:16.994Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/docs/LOCAL_APPLIANCE_ARCHITECTURE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

43. user (2026-07-02T04:56:20.953Z)

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44. user (2026-07-02T04:56:33.327Z)

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45. user (2026-07-02T04:56:40.600Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-droplet-docrot/docs/DRY_RUN_CUJ.md has been updated successfully. (file state is current in your context ? no need to Read it back)

46. user (2026-07-02T04:56:44.840Z)

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47. user (2026-07-02T04:56:48.258Z)

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48. assistant (2026-07-02T04:56:58.031Z)

All edits are in. Now verify: remaining stale references, other citations that may have drifted, and whether CI has any doc checks.

49. user (2026-07-02T04:57:00.337Z)

No matches found

50. user (2026-07-02T04:57:02.390Z)

```
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-docrot\deploy\GPU_VM_SERVING.md:4:> runbook treats as the current host / "hot fallback" was
**decommissioned + deleted 2026-06-25**
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\scratchpad\wt-droplet-docrot\deploy\GPU_VM_SERVING.md:38:> cutover with the droplet as a hot fallback ? that box no longer exists, so there is no fallback to
```

51. user (2026-07-02T04:57:03.699Z)

```
name: test
on:
  push:
    branches: [master]
  pull_request:

jobs:
  site:
    runs-on: ubuntu-latest
    defaults:
      run:
        working-directory: site
    steps:
      - uses: actions/checkout@v7
      - uses: actions/setup-node@v6
        with:
          node-version: "24" # node:sqlite is unflagged; server.test.js boots the VM host
          cache: "npm"
          cache-dependency-path: site/package-lock.json
      # jsdom is a TEST-ONLY devDep (the behavioral landing-page UX tests run the real inline
      # script in a DOM); the Cloudflare Worker + VM-host RUNTIMES stay dependency-free. `npm
test`
the
  # runs ALL paths: Worker+D1 logic (index.test.js), the booted VM host (server.test.js),
  # static content contract (assets.test.js), and the UX behaviour (ux.dom.test.js).
  - run: npm ci
  - run: npm test
  # Syntax-check the ops shell scripts (provision/teardown/smoke/parity) so a broken
  # script can't ship to the runbook. `bash -n` parses without executing.
  - name: shell scripts parse-check
    run: for f in scripts/*.sh; do echo "checking $f"; bash -n "$f"; done

web:
  # Operator-console SPA (web/): typecheck + unit tests (with a coverage floor) + production
build
  # gate the merge. `test:coverage` fails if coverage drops below the thresholds in
```

```

vite.config.ts,
# so a merge can't silently regress test coverage.
runs-on: ubuntu-latest
defaults:
  run:
    working-directory: web
steps:
  - uses: actions/checkout@v7
  - uses: actions/setup-node@v6
    with:
      node-version: "24"
      cache: "npm"
      cache-dependency-path: web/package-lock.json
  - run: npm ci
  - run: npm run typecheck
  - run: npm run test:coverage
  - run: npm run build

layout:
# Real-browser layout-regression guard. The other suites run in jsdom, which has NO layout
engine
# (getBoundingClientRect() is all zeros) and therefore cannot see header overflow / wrapping
/
# clipping ? the responsive-nav bug that recurred. This drives a headless Chromium over the
real
# pages at a spread of viewport widths x all 6 locales and asserts the header never
distorts.
runs-on: ubuntu-latest
defaults:
  run:
    working-directory: site
steps:
  - uses: actions/checkout@v7
  - uses: actions/setup-node@v6
    with:
      node-version: "24"
      cache: "npm"
      cache-dependency-path: site/package-lock.json
  - run: npm ci
  - run: npx playwright install --with-deps chromium
  - run: npm run test:layout
# Cross-locale render guard: drives the real pages in every locale and asserts the live
DOM
# matches the catalog (no raw keys / no shim regression / right lang / no console errors).
- run: npm run test:render
# Accessibility guard: axe-core WCAG 2.1 A/AA over every page x representative locales;
fails on
# any serious/critical violation (jsdom has no contrast/layout engine to catch these).
- run: npm run test:ally

```

52. user (2026-07-02T04:57:18.529Z)

```

M NEXT_STEPS.md
M deploy/DEPLOY_ALPHA.md
M deploy/GPU_VM_SERVING.md
M deploy/droplet/README.md
M docs/DRY_RUN_CUJ.md
M docs/LOCAL_APPLIANCE_ARCHITECTURE.md
docs/retire-droplet-doc-rot
f9eea14 web: resilient video-summary readers + show fps in the library (#140)

```

53. user (2026-07-02T04:57:25.250Z)

```

NEXT_STEPS.md | 35 ++++++-----
deploy/DEPLOY_ALPHA.md | 24 ++++++---

```

```

deploy/GPU_VM_SERVING.md          | 96 ++++++-----
deploy/droplet/README.md          | 10 ++++
docs/DRY_RUN_CUJ.md               | 6 +-
docs/LOCAL_APPLIANCE_ARCHITECTURE.md | 10 +--
6 files changed, 117 insertions(+), 64 deletions(-)
diff --git a/NEXT_STEPS.md b/NEXT_STEPS.md
index 28e1b99..9d2a219 100644
--- a/NEXT_STEPS.md
+++ b/NEXT_STEPS.md
@@ -2,17 +2,19 @@

```

Prioritized "what to do next" for the **khelsutra** product repo. Refresh at each checkpoint.

```

-Where we are (updated 2026-07-01): khelsutra.guru is LIVE and fully shipped ? landing
page + `/capture`
+Where we are (updated 2026-07-02): khelsutra.guru is LIVE and fully shipped ? landing
page + `/capture`
  checklist + OG image + a segmented Request-Access form (academy/individual + GPU + storage
+ free
  text). On Cloudflare (Worker `hidden-king-khelsutra` + D1 `khelsutra-access`): the form
persists to
  D1 and emails a summary to the owner on every registration (Email Routing send binding ?
verified
-live). Git auto-deploy is connected (Workers Builds, root `site/`): merge to `master` ?
deploys;
-CI gates every PR ? three jobs: the full site `npm test` (Worker + VM host + assets + UX),
the
-`web/` SPA build + coverage floor, and a real-browser layout check. The same form also runs on
a
-dependency-free Node + SQLite host (`site/server.js`), live + browsable on the D0 droplet
?
-http://168.144.126.176:8787 (systemd `khelsutra-site`). Shipped across ~90 merged PRs (the
form,
-6-language localization, the per-rally MVP, droplet portability). Docs:
-`docs/REQUEST_ACCESS_FORM.md`, `docs/PORTABILITY.md`.
+live). Cloudflare is the single canonical host ? the D0 droplet mirror (`168.144.126.176`)
was
+decommissioned + deleted 2026-06-25 (#117; data archived to
+gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/). The dependency-
free
+Node + SQLite host (`site/server.js`) remains VM-portable (live-proven on the droplet
before its
+retirement) but nothing is deployed on it today. Git auto-deploy is connected (Workers
Builds, root
+`site/`): merge to `master` ? deploys; CI gates every PR ? three jobs: the full site `npm
test`
+(Worker + VM host + assets + UX), the `web/` SPA build + coverage floor, and a real-browser
layout
+check. Shipped across ~90 merged PRs (the form, 6-language localization, the per-rally MVP, VM
+portability). Docs: `docs/REQUEST_ACCESS_FORM.md`, `docs/PORTABILITY.md`.

```

Next product threads (owner ideas, 2026-06-20)

```

- ? DONE ? Friendly & Multilingual (ARCHIVED 2026-06-22; human gates cleared): the product
is
@@ -58,10 +60,15 @@ http://168.144.126.176:8787 (systemd `khelsutra-site`). Shipped across ~90
mer
  (Cloudflare can't rewrite that ? no injection), and after a deploy the rendered HTML should
contain the
  edge-injected beacon. DONE WHEN: the Web Analytics dashboard shows incrementing page
views for
  `khelsutra.guru`.
-- Droplet ? production ? TLS + a subdomain (`vm.khelsutra.guru` via Caddy) if it should be
a real
- standby; add VM-path SMTP so the Node host also emails (currently `noopNotify`).
+- Any future VM ? production *(re-scoped 2026-07-02 ? was "Droplet ? production"; the

```

```

droplet was
+ deleted 2026-06-25, #117)* ? only if a VM standby is ever wanted again:
`site/scripts/provision-vm.sh`
+ is provider-agnostic; add TLS + a subdomain (`vm.khelsutra.guru` via Caddy) and **VM-path
SMTP** so the
+ Node host also emails (currently `noopNotify`). Not urgent ? Cloudflare is the single
canonical host.
+- **Revoke the droplet SSH key (`~/ssh/khelsutra_droplet`) ? still actionable, do it now
(owner):**
+ the droplet is gone (deleted 2026-06-25) but the local keypair lingers; revoke/delete it
rather than
+ waiting on any VM work.
- **DMARC** TXT for deliverability; delete any orphan `khelsutra-site` Worker; clean CF D1 test
rows
- (`deploy-test@`, `email-test@`). Revoke the droplet SSH key (`~/ssh/khelsutra_droplet`) when
done.
+ (`deploy-test@`, `email-test@`).

## Bigger arcs (design-first)
- **? Studio media lifecycle (ingest ? store ? annotate ? train)** ? **Tracked project:
`docs/STUDIO_MEDIA_LIFECYCLE.md`
@@ -74,9 +81,9 @@ http://168.144.126.176:8787** (systemd `khelsutra-site`). Shipped across ~90
mer
- **Single-box appliance (KhelSutra Studio)** ? **? Tracked design:
`docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (DRAFT, owner-gated).**
The `web/` SPA drives the full **ingest (Drive / bucket URI / local upload) ? index ? pick
rallies ? reel ? download**
loop on ONE box, behind an **edge auth gate** (engine stays `127.0.0.1`, CORS `` is moot).
It is the single-box
- profile of the engine's decoupled `StorageBackend` + `backend.worker` architecture. MVP runs
on the CPU droplet via
- **Gemini/stub**; **real local-GPU detection needs a GPU host** (DO droplet / GCE VM / Cloud
Run+GPU ? owner priority
- ASAP, runs parallel to the UI MVP). Per-rally selection = one screen. Owner decisions pending
in $10 (edge auth,
+ profile of the engine's decoupled `StorageBackend` + `backend.worker` architecture. MVP ran
on the CPU droplet
+ (retired 2026-06-25) via **Gemini/stub**; **real local-GPU detection needs a GPU host** (any
VM provider / Cloud
+ Run+GPU ? owner priority ASAP, runs parallel to the UI MVP). Per-rally selection = one
screen. Owner decisions pending in $10 (edge auth,
retention, cost ceiling, GPU host, P5 bridge).
- **Alpha SPA** in `web/` (Vite + React + TS + Tailwind) ? hosts the operator console + the
per-rally selection UI above.
- **BYO-storage + autosync** ? connect Drive/OneDrive/GCS/S3, watch the folder, auto-prepare
reels
diff --git a/deploy/DEPLOY_ALPHA.md b/deploy/DEPLOY_ALPHA.md
index 59b3238..11f754b 100644
--- a/deploy/DEPLOY_ALPHA.md
+++ b/deploy/DEPLOY_ALPHA.md
@@ -1,12 +1,19 @@
# Deploy KhelSutra Studio for alpha ? a shareable page, no SSH

-> **Status:** `LIVE` (deploy live-verified on the droplet 2026-06-22 ?
`deploy/droplet/README.md`; only a rotated Gemini key + a browser OTP login remain to exercise
the upload?reel CUJ) . **Topology:** split ? SPA on Cloudflare Pages + engine on an `api.*`
cloudflared tunnel, Cloudflare Access over both (the engine's `HOSTING_PLAN.md` $2 alpha plan).
. **Cost:** ~$0/mo standing (Pages + Tunnel + Access are free ?50 users) + Gemini per-run
(~$0.05/match).
+> **Status:** `RETIRED / HISTORICAL` (updated 2026-07-02). The alpha this runbook stood up went
live on the DO droplet `168.144.126.176` on 2026-06-22 (`deploy/droplet/README.md`), and that
droplet was **decommissioned + deleted 2026-06-25** (PR
[#117](https://github.com/avidullu/khelsutra/pull/117), commit `5ffb750`; its data archived to
`gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/`). Alpha serving is
**paused pending the Cloud-Run move**. The Cloudflare/engine mechanics below remain the

```

reference for the next stand-up, but every "droplet" reference is historical ? do ****not**** deploy to or SSH that IP (DigitalOcean can recycle it).

+>
+> ****Durable topology decision ? single-origin is the DEFAULT for SPA + CF-Access.**** The ****split**** topology this runbook originally shipped (SPA on Cloudflare Pages + engine on an `api.` cloudflared tunnel, Access over both) broke Cloudflare Access in practice ? the cross-origin split defeated the Access session/preflight flow and locked testers out (issues [#97](https://github.com/avidullu/khelsutra/issues/97), [#114](https://github.com/avidullu/khelsutra/issues/114)); the 2026-06-22 fix repointed serving to a ****single origin**** (engine-served SPA). Any future stand-up should serve the SPA and the API from ONE origin (`docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md` D1 / the \$0 "Alternative (single-origin)" note below). This paragraph records the decision so #97 can be closed.

+>
+> ****Still pending (owner, Cloudflare dashboard):**** clean up the dangling `app.`/`api.khelsutra.guru` CNAMEs and the CF-Access application ? they still resolve via Cloudflare to a tunnel with no origin.

+>
+> ****As-shipped facts (historical): **Topology:**** split (superseded ? see above) . ****Cost:**** ~\$0/mo standing (Pages + Tunnel + Access free ?50 users) + Gemini per-run (~\$0.05/match).*

This turns the local-only flow (run engine + SPA on your box, drive it yourself) into a ****page** you can share with alpha testers** ? they open `app.khelsutra.guru`, sign in (email allowlist), point at a video, and get a reel. No SSH, no home IP exposed, no inbound ports.

-> ****Outgrown the droplet?**** To move the live engine onto a ****GCP L4 GPU VM**** (more CPU/RAM for ffmpeg

-> + optional WASB GPU detection) and cut the tunnel over to it, see

****GPU_VM_SERVING.md****

-> ? same tunnel + Access, no DNS change, droplet kept as fallback (~\$500?600/mo while the VM runs).

+> ****Need more compute?**** To run the engine on a ****GCP L4 GPU VM**** (more CPU/RAM for ffmpeg

+> + optional WASB GPU detection) behind the same tunnel + Access, see

****GPU_VM_SERVING.md****

+> ? no DNS change (~\$500?600/mo while the VM runs). *Its "droplet kept as fallback" language is

+> historical too ? there is no droplet anymore.*

0. The shape

@@ -26,13 +33,13 @@ This turns the local-only flow (run engine + SPA on your box, drive it yourself)

- ****Auth at the edge.**** Cloudflare Access gates both hostnames; the engine **also** verifies the `Cf-Access` JWT in-app (defence in depth ? `security.edge\_auth\_enabled`), so a direct hit to the tunnel can't spoof identity.

- ****Cross-origin, on purpose.**** `app.` (SPA) and `api.` (engine) are different origins ? the SPA is built with `VITE\_API\_BASE=https://api.khelsutra.guru/api` (PR #64) and the engine locks CORS to the SPA origin via `RALLY\_CORS\_ALLOW\_ORIGINS` (no engine code change ? it's already env-driven).

-> ****Alternative (single-origin):**** if you'd rather run ****one box**** that serves the SPA **and** the API behind one origin (Caddy reverse-proxy, no Pages, no cross-origin), see `LOCAL\_APPLIANCE\_ARCHITECTURE.md` D1. This runbook ships the ****split**** topology you chose (stable Pages-hosted page + a thin API tunnel).

+> ****Single-origin is now the DEFAULT** (decision recorded 2026-07-02 ? see the status banner / issue #97): ****run one box**** that serves the SPA **and** the API behind one origin (engine-served SPA, or a Caddy reverse-proxy ? `docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md` D1), so Cloudflare Access sees a single origin. The ****split**** topology this runbook originally shipped (stable Pages-hosted page + a thin API tunnel) is kept below for reference but is ****superseded**** ? it broke the Access sign-in flow (#97, #114).

1. Prerequisites

-- A ****box**** to run the engine: the CPU droplet `168.144.126.176` (Gemini-only ? no GPU; fine for alpha) ****or**** your Windows+WSL GPU box (real WASB). `ffmpeg` + Python 3.8+ installed; the

```

`badminton-highlight-indexer` engine checked out.
+- A box to run the engine: any Linux VM (Gemini-only is fine for alpha ? the retired stand-
up used a CPU DO droplet, deleted 2026-06-25) or your Windows+WSL GPU box (real WASB).
`ffmpeg` + Python 3.8+ installed; the `badminton-highlight-indexer` engine checked out.
- A Cloudflare account with the `khealsutra.guru` zone (you already run the marketing site
there).
- `cloudflared` installed on the box (`https://developers.cloudflare.com/cloudflare-
one/connections/connect-networks/downloads/`).
- A fresh `GEMINI_API_KEY` (rotate the chat-shared one ? it uploads downsized chunks to
Google; this is a paid + cloud action).
@@ -79,7 +86,7 @@ export RALLY_BIND_HOST="127.0.0.1" # keep the
loopba
```bash
python -m backend.main --server # binds 127.0.0.1:8000 ? kept loopback by RALLY_BIND_HOST
($2.2); with edge_auth ON and that var unset it would bind 0.0.0.0 instead
```

-Keep it alive with a process manager (`systemd` on the droplet, mirroring
`site/scripts/provision-vm.sh`'s unit; or `pm2`/`nssm` on Windows).
+Keep it alive with a process manager (`systemd` on a Linux VM, mirroring
`site/scripts/provision-vm.sh`'s unit; or `pm2`/`nssm` on Windows).

---

@@ -140,7 +147,7 @@ See `deploy/cloudflared.config.example.yml` for the ingress (only
`api.khealsutra`
- You MUST set `edge_auth_enabled=true` to expose. The boot posture (and the in-app JWT
verify) only engage then. Do not put the engine behind the tunnel with edge auth off.
- Cross-origin reel download: the reel streams from `api.` with `Content-Disposition:
attachment`, so it downloads even cross-origin ? but verify the filename in your browser during
the CUJ (cross-origin `` filename hints are ignored; the header is the authority).
- Upload size: Cloudflare's free proxy caps request bodies at 100 MB ? large matches
upload as ~95 MB chunks (the SPA chunked-upload path honours the engine's `max_part_mb`, default
95; lower `security.max_part_mb` to shrink them). `gs://` bucket pre-staging is not an
option on this box (see the jail caveat above).
-- Box offline = engine offline. The Pages SPA stays up and shows a friendly "engine
offline" banner (`/api/health` ping); processing resumes when the box is back. Beta removes this
with a queue (`HOSTING_PLAN.md` §3).
+- Box offline = engine offline. The Pages SPA stays up and degrades gracefully ? today that
means the SetupWizard's offline branch (driven by its `/api/capabilities` fetch), not a
persistent "engine offline" banner; the SPA has no `/api/health` ping (unimplemented ? corrected
2026-07-02). Processing resumes when the box is back. Beta removes this with a queue
(`HOSTING_PLAN.md` §3).
- Gemini = paid + uploads footage to Google. Keep the key rotated; use it only for clips
you're OK sending.

---

@@ -150,5 +157,6 @@ See `deploy/cloudflared.config.example.yml` for the ingress (only
`api.khealsutra`

---

### Changelog
+- 2026-07-02 ? RETIRED / HISTORICAL. The droplet this runbook deployed to was
decommissioned + deleted 2026-06-25 (PR #117; data archived to `gs://khealsutra-rally-
corpus/archive/droplet-claude-do-rally-test-20260625/`) ? no `.md` recorded that until now.
Recorded the durable single-origin-default topology decision for SPA + CF-Access (#97,
#114); re-scoped the §1 box prerequisite to "any Linux VM"; softened the unimplemented
`/api/health` "engine offline banner" claim (§7); flagged the CF-dashboard CNAME/Access cleanup
as pending owner click-ops.
- 2026-06-22 ? correctness pass (adversarial verify against engine `origin/master` + a
hermetic repro). Fixed two would-break-the-deploy gaps: (1) RALLY_ALLOWED_HOSTS was
missing ? with `edge_auth` on, the Host-header guard 400s every tunneled request until it's
set (added it + `RALLY_BIND_HOST` to §2.2; repro confirmed 400?200); (2) gs://` ingest is
rejected by the `allowed_video_root` jail on the exposed box (dropped the `gs://` promise from
§6/§7; alpha path = browser upload). Minor: 401-not-500 wording (§2.1), `config.json`
location/`--setup` note (§2.2), bind comment (§2.3), ~95 MB chunk size (§7).

```

```

- 2026-06-21 ? created. Split-topology alpha deploy kit (cloudflared + Pages + CF Access)
consuming the engine exposure-safety boot posture (engine #294) + the SPA `VITE_API_BASE` (#64).
Pairs with `LOCAL_APPLIANCE_ARCHITECTURE.md` (single-origin alternative) + the engine
`HOSTING_PLAN.md` §2.
diff --git a/deploy/GPU_VM_SERVING.md b/deploy/GPU_VM_SERVING.md
index 70ec872..65cc60e 100644
--- a/deploy/GPU_VM_SERVING.md
+++ b/deploy/GPU_VM_SERVING.md
@@ -1,12 +1,24 @@
# Move KhelSutra Studio onto an L4 GPU VM (off the droplet)

+> *** Droplet references in this doc are HISTORICAL (noted 2026-07-02).*** The DO CPU droplet
this
+> runbook treats as the current host / "hot fallback" was **decommissioned + deleted
2026-06-25**
+> (PR [#117](https://github.com/avidullu/khelsutra/pull/117)); data archived to
+> `gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/`. Nothing is
serving
+> today (alpha paused pending the Cloud-Run move). The VM mechanics (§1?§5, §9) remain valid;
the
+> cutover/rollback steps that SSH the droplet (§6, §8) have no live box to talk to ? source the
+> secrets from their **canonical stores** (§0) and simply run the connector on the VM. Also
note the
+> durable decision since: **single-origin is the default topology for SPA + CF-Access**
+> ([`DEPLOY_ALPHA.md`](DEPLOY_ALPHA.md) status banner / issue #97) ? a fresh stand-up should
serve the
+> SPA and API from one origin rather than reviving the split `app.`/`api.` shape assumed below.
+
Self-serve runbook to run the live engine (behind `api./app.khelsutra.guru`) on a **GCP L4 GPU
VM**
instead of the CPU droplet, and **cut the Cloudflare tunnel over** so traffic lands on the VM.
Lets
you switch detection between **Gemini** (cloud API, proven) and **WASB native** (uses the L4
GPU).

-This extends [`DEPLOY_ALPHA.md`](DEPLOY_ALPHA.md) (which deployed the same engine on the
droplet) ? it
-reuses the droplet config templates in [`deploy/droplet/`](droplet/) and the GPU-VM scripts in
the
-engine repo (`badminton-highlight-indexer`: `deploy/gcp/`).
+This extends [`DEPLOY_ALPHA.md`](DEPLOY_ALPHA.md) (which deployed the same engine on the now-
retired
+droplet) ? it reuses the droplet config templates in [`deploy/droplet/`](droplet/) (historical
+templates, still valid as templates) and the GPU-VM scripts in the engine repo
+(`badminton-highlight-indexer`: `deploy/gcp/`).

> These are manual steps today. Automating them with a one-shot startup-script you feed
into
> the GCP console (`create_gpu_vm.sh` already injects a metadata startup-script ? today it only
installs
@@ -20,10 +32,11 @@ reuses the droplet config templates in [`deploy/droplet/`](droplet/) and the
GPU
> Cloud Run GPU service would be far cheaper (~$0 idle) ? that's the backlogged "M7" and
a bigger
> build; this VM is the fast path that works today.
>
-> Cutover is safe + reversible. Bring the VM up and verify it serving before touching
the
-> tunnel; the droplet stays as a hot fallback. The tunnel is token-managed (ingress lives
in the
-> Cloudflare dashboard, not on the box), so moving it is just "run the same connector on the
VM, then
-> stop it on the droplet" ? no DNS change, no new Cloudflare token, no CF-Access change.
+> Cutover is simple. The tunnel is token-managed (ingress lives in the Cloudflare
dashboard,

```

```
+> not on the box), so serving from the VM is just "run the connector on the VM" ? **no DNS
change, no
+> new Cloudflare token, no CF-Access change.** *(Historical: this originally described a zero-
downtime
+> cutover with the droplet as a hot fallback ? that box no longer exists, so there is no
fallback to
+> keep warm and nothing to stop afterwards.)*
```

@@ -32,12 +45,20 @@ reuses the droplet config templates in [`deploy/droplet/`](droplet/) and the GPU

```
- **gcloud** authenticated as the project owner (gcloud auth login) on the machine you run
this from.
- **GPU quota** GPUS_ALL_REGIONS ? 1 and the SA Ops-Agent roles ? see the engine repo
deploy/gcp/README.md §2. (You already have L4 quota=1 from the nightly runs.)
-- **The two secrets**, which you already have from the droplet deploy:
- - the **rotated Gemini API key** (Drive: gemini.api_key.txt) ? only needed for the Gemini
path;
- - the **CF-Access AUD** + team domain ? already in the droplet's /srv/khelsutra/config.json
- (security.cf_access_aud / cf_team_domain); reuse the same values.
- - the **tunnel token** ? in the droplet's cloudflared unit (systemctl cat cloudflared ? the
- --token ? value). You'll copy it to the VM in §6.
+- **The secrets ? from their canonical stores** (do NOT try to read them off the droplet; it
was
+ deleted 2026-06-25):
+ - the **rotated Gemini API key** ? the owner's secret store (Drive: gemini.api_key.txt);
only
+ needed for the Gemini path;
+ - the **CF-Access AUD** + team domain ? Cloudflare dashboard: **Zero Trust ? Access ?
Applications ?
+ the khelsutra app ? its AUD tag*; team domain is https://<your-
team>.cloudflareaccess.com.
+ (The retired droplet's /srv/khelsutra/config.json ? with security.cf_access_aud /
+ cf_team_domain filled in ? is also preserved in the GCS archive
+ gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/ if you need
the
+ as-deployed values.)
+ - the **tunnel token** ? Cloudflare dashboard: **Zero Trust ? Networks ? Tunnels ?
+ khelsutra-studio ? Configure** (the tunnel is token-managed, so the dashboard shows the
+ connector install command containing the token, and can regenerate it). You'll use it on
the VM
+ in §6.
- **For the WASB path only:** the weights at gs://khelsutra-rally-
corpus/models/wasb_badminton_best.pth.tar.
```

You'll work in **two repo checkouts** on your machine: this one (`khelsutra`) for the config templates,

@@ -119,8 +140,9 @@ The native runner needs **three** env vars (set them in `engine.env`, §4):
`WAS`

4. Serving config + start

```
-Reuse the droplet templates from this repo's [deploy/droplet/](droplet/) ? the **AUD, team
domain,
-and hosts are identical** to the droplet (it's the same tunnel + Access app).
+Reuse the (historical) droplet templates from this repo's [deploy/droplet/](droplet/) ? the
**AUD,
+team domain, and hosts are the same values** the retired droplet deploy used (same tunnel +
Access app;
+get the actual values per §0).
```

```
**/srv/khelsutra/config.json** (from config.json.example): keep
security.edge_auth_enabled=true,
allowed_video_root=/srv/khelsutra/incoming, your real cf_team_domain + cf_access_aud. Set
```

@@ -175,8 +197,9 @@ The server default is one config flip + restart. (API callers can also override

Apply: edit `config.json`, then `sudo systemctl restart khelsutra-engine`. Confirm with `curl -s localhost:8000/api/capabilities` (`default\_segmenter`) and a real run.

-> Start on **Gemini** (it's exactly what's live now), prove the VM serves end-to-end, cut the tunnel

-> over (\$6), then flip to **WASB** once you're happy ? so you change one variable at a time.

+> Start on **Gemini** (the proven serving path ? it's what the retired droplet alpha ran), prove the VM

+> serves end-to-end, cut the tunnel over (\$6), then flip to **WASB** once you're happy ? so you change

+> one variable at a time.

@@ -186,28 +209,28 @@ The tunnel `khelsutra-studio` is **token-managed**: its ingress (``api.khelsutra``

lives in the Cloudflare dashboard and follows whichever machine runs the connector. So:

```bash

-# 1. get the token from the droplet (it's your secret; don't paste it anywhere public)

-ssh root@<droplet-ip> 'systemctl cat cloudflared' | grep -o -- '--token [A-Za-z0-9.\_=-]'

+# 1. get the token from the Cloudflare dashboard (Zero Trust ? Networks ? Tunnels ? khelsutra-studio

+# ? Configure ? the connector install command contains the token; see \$0). It's your secret; don't

+# paste it anywhere public.

+# (Historical: this step used to read the token off the droplet via `systemctl cat cloudflared` ?

+# that box was deleted 2026-06-25.)

# 2. on the VM: install cloudflared + run the SAME token

gcloud compute ssh khelsutra-gpu --zone=\$ZONE

curl -fsSL https://pkg.cloudflare.com/cloudflared-stable-linux-amd64.deb -o /tmp/cf.deb &&  
sudo dpkg -i /tmp/cf.deb

sudo cloudflared service install <THE\_TOKEN> # installs + starts the connector as a service

- sudo systemctl status cloudflared # "Registered tunnel connection" ? the VM is now a replica

+ sudo systemctl status cloudflared # "Registered tunnel connection" ? the VM serves the tunnel

#### 54. assistant (2026-07-02T04:57:29.701Z)

The diff looks good so far. Let me review the remainder.

#### 55. user (2026-07-02T04:57:32.381Z)

-# 1. get the token from the droplet (it's your secret; don't paste it anywhere public)

-ssh root@<droplet-ip> 'systemctl cat cloudflared' | grep -o -- '--token [A-Za-z0-9.\_=-]'

+# 1. get the token from the Cloudflare dashboard (Zero Trust ? Networks ? Tunnels ? khelsutra-studio

+# ? Configure ? the connector install command contains the token; see \$0). It's your secret; don't

+# paste it anywhere public.

+# (Historical: this step used to read the token off the droplet via `systemctl cat cloudflared` ?

+# that box was deleted 2026-06-25.)

# 2. on the VM: install cloudflared + run the SAME token

gcloud compute ssh khelsutra-gpu --zone=\$ZONE

curl -fsSL https://pkg.cloudflare.com/cloudflared-stable-linux-amd64.deb -o /tmp/cf.deb &&  
sudo dpkg -i /tmp/cf.deb

sudo cloudflared service install <THE\_TOKEN> # installs + starts the connector as a

```

service
- sudo systemctl status cloudflared # "Registered tunnel connection" ? the VM is
now a replica
+ sudo systemctl status cloudflared # "Registered tunnel connection" ? the VM
serves the tunnel
```

```

-At this point **both** the droplet and the VM serve the tunnel (Cloudflare HA ? either can answer; both
 -run the same engine, so that's fine). Verify the VM is healthy through the public hostname:
 +With the droplet gone, the VM is the tunnel's **only** connector. Verify it's healthy through the
 the
 +public hostname:

```

```bash
curl -sI https://api.khelsutra.guru/api/health # 200 (or the 302?Access login if
unauthenticated ? that's correct)
```

```

-Then **stop the droplet's connector** so all traffic lands on the VM:

```

-
-```bash
-ssh root@<droplet-ip> 'systemctl stop cloudflared && systemctl disable cloudflared'
-```

```

*(Historical: the original cutover ran droplet + VM as HA replicas, then stopped the droplet's
 +connector via SSH. There is no droplet connector to stop anymore.)*

DNS, the CF-Access app, and the dashboard ingress are unchanged ? only the connector host
 moved.

@@ -224,15 +247,17 @@ DNS, the CF-Access app, and the dashboard ingress are unchanged ? only the
 con

8. Rollback (if anything's off)

-The droplet is untouched (engine still installed, just its cloudflared stopped):
 +> **HISTORICAL** ? no droplet fallback exists.** The rollback below assumed the droplet was still
 +> installed with its connector merely stopped; that box was deleted 2026-06-25. Today
 "rollback" is
 +> simply stopping the VM's connector (`sudo systemctl stop cloudflared` on the VM) ? the tunnel
 then
 +> has no origin and the app is offline, which is the current steady state anyway (alpha
 paused).

```

```bash
-ssh root@<droplet-ip> 'systemctl enable --now cloudflared' # droplet serves again
-gcloud compute ssh khelsutra-gpu --zone=$ZONE --command 'sudo systemctl stop cloudflared' #
VM stops serving
+ (dead steps, kept for the record ? <droplet-ip> no longer exists)
+ ssh root@<droplet-ip> 'systemctl enable --now cloudflared' # droplet serves again
+ gcloud compute ssh khelsutra-gpu --zone=$ZONE --command 'sudo systemctl stop cloudflared' #
VM stops serving
```

```

-(Run the droplet start **before** stopping the VM for zero-gap rollback.)

```

-
---
```

9. Cost control / teardown

@@ -251,6 +276,7 @@ gcloud compute ssh khelsutra-gpu --zone=\$ZONE --command 'sudo systemctl stop
 clo

```

|---|---|
| Provision / teardown the L4 VM | engine repo `deploy/gcp/{create_gpu_vm.sh,teardown.sh}` |
| WASB conda env builder | engine repo `deploy/digitalocean/gpu_setup.sh` |
-| Serving config templates | this repo
```

```

`deploy/droplet/{config.json.example,engine.env.example,khelsutra-engine.service}` |
-| Tunnel ingress (managed in CF dashboard) | Zero Trust ? Networks ? Tunnels ? `khelsutra-
studio` |
-| The original droplet deploy | [`DEPLOY_ALPHA.md`](DEPLOY_ALPHA.md) |
+| Serving config templates (historical droplet templates, still valid) | this repo
`deploy/droplet/{config.json.example,engine.env.example,khelsutra-engine.service}` |
+| Tunnel ingress + token (managed in CF dashboard) | Zero Trust ? Networks ? Tunnels ?
`khelsutra-studio` |
+| The original droplet deploy (RETIRED 2026-06-25) | [`DEPLOY_ALPHA.md`](DEPLOY_ALPHA.md) |
+| The droplet's archived data (incl. its `config.json`) | `gs://khelsutra-rally-
corpus/archive/droplet-claude-do-rally-test-20260625/` |
diff --git a/deploy/droplet/README.md b/deploy/droplet/README.md
index 861f710..deb6280 100644
--- a/deploy/droplet/README.md
+++ b/deploy/droplet/README.md
@@ -1,5 +1,15 @@
# Droplet quickstart ? KhelSutra alpha (the `api.` engine side)

+> RETIRE ? this droplet no longer exists (banner added 2026-07-02). The DO droplet
+> `168.144.126.176` was decommissioned and deleted 2026-06-25 (PR
+> [#117](https://github.com/avidullu/khelsutra/pull/117), commit `5ffb750`; its data archived
to
+> `gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/`). Do NOT SSH
or
+> deploy to that IP ? DigitalOcean can recycle it to a stranger. This file is kept as a
historical
+> template for a future Linux-VM stand-up (the config/env/systemd templates alongside it are
still
+> valid as templates). Read `../DEPLOY_ALPHA.md` (also RETIRED/HISTORICAL) first ? including
the
+> durable decision recorded there: single-origin is the default topology for SPA + CF-
Access
+> (issues #97/#114), so don't revive this split `app.`/`api.` shape as-is.
+
The filled-in, command-dense version of `../DEPLOY_ALPHA.md` for the Linux CPU droplet
(`168.144.126.176`, Gemini, no GPU). Values are pre-baked for app.khelsutra.guru` /
`api.khelsutra.guru`, team domain https://khelsutra.cloudflareaccess.com, allowlist
diff --git a/docs/DRY_RUN_CUJ.md b/docs/DRY_RUN_CUJ.md
index 36a8d5b..24568c3 100644
--- a/docs/DRY_RUN_CUJ.md
+++ b/docs/DRY_RUN_CUJ.md
@@ -27,7 +27,7 @@
@@ -27,7 +27,7 @@ Operate the web UI ? point at a video on the GCS bucket ? the engine
Path B ? Full "point at a bucket video IN the UI" (SPA-only ? smaller than first scoped).
The HTTP bucket-ingest endpoint already exists ($1) ? so Path B is only the SPA front-
half (appliance S1/S2): an ingest input (enter a `gs://` URI) ? `POST /api/process` ? an
ingest + progress screen polling `GET /api/jobs/{id}` ? on `done`, load `result.video_id`
(which the existing back-half already renders). No net-new engine endpoint ? B-P1 is already
done; the remaining work is B-P2 (the SPA).
```

3. Path A runbook (the cheapest dry run) `[verified-steps]`

- Engine host + deps:** run the engine FastAPI (`python -m backend.main --server`, binds
`127.0.0.1:8000`) on a box with `ffmpeg` + OpenCV. Gemini/motion paths are **CPU-only**, no WSL ?
local Windows works; the CPU droplet works too. (GPU/WSL only needed for native WASB.)
- Engine host + deps:** run the engine FastAPI (`python -m backend.main --server`, binds
`127.0.0.1:8000`) on a box with `ffmpeg` + OpenCV. Gemini/motion paths are **CPU-only**, no WSL ?
local Windows works; any Linux CPU box works too (the old CPU droplet is gone ? deleted
2026-06-25). (GPU/WSL only needed for native WASB.)
- Get the bucket video local + into the API DB:**

```

gsutil cp gs://<bucket>/<match>.mp4 ./incoming/ (or the worker fetch), then `POST
/api/process {"video_path": "<abs local path>", "segmenter": "gemini"}` ? poll `GET
/api/jobs/{job_id}` until `result.video_id`. *(`gemini` needs `GEMINI_API_KEY`; `motion` is key-
free but hides fabricated metadata.)*

```
- Open the SPA against the engine:** `npm run dev` (Vite proxies `/api` ? `:8000`, same-
origin) ? `http://localhost:5173/?video=<video\_id>` ? rallies auto-load.

```

@@ -39,7 +39,7 @@
@@ -39,7 +39,7 @@ Operate the web UI ? point at a video on the GCS bucket ? the engine

```

```

## 4. Open decisions ? OWNER (these gate the dry run)
1. **Scope:** Path A (cheapest dry run now) or Path B (the in-UI bucket ingest ? **SPA-only**);
the engine endpoint already accepts `gs://`, see §1)? *Recommend A first* (proves the
multilingual workflow fast), then B as the appliance build. ? ? **RESOLVED: A first (done, §7),
then B (B-P2, the SPA ingest/progress screen).**
2. **Compute / detector:** **Gemini** (real rallies; needs your `GEMINI_API_KEY`; uploads
downsized chunks to Google = paid + cloud) . **motion** (key-free, runs anywhere, but lower-
quality boundaries + fabricated server/shot metadata stays hidden) . **GPU WASB** (best quality,
but needs a GPU host + is owner-parity-gated, not production-verified). *Recommend Gemini for
"real-ish, cheapest" ? pending your key + OK (paid + uploads footage).*
-3. **Engine host:** local Windows (simplest, Gemini/motion) . the CPU droplet `168.144.126.176`
(shared testbed ? careful) . a GPU VM (only for WASB).
+3. **Engine host:** local Windows (simplest, Gemini/motion) . ~~the CPU droplet
`168.144.126.176` (shared testbed ? careful)~~ *(retired ? deleted 2026-06-25, PR #117)* . a GPU
VM (only for WASB).
4. **Which bucket video:** e.g. `gs://khesutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4`
(already staged, known-good) or another clean ?720p clip (must pass the engine validators).

```

```

## 5. Honest caveats
@@ -53,7 +53,7 @@ Operate the **web UI** ? point at a **video on the GCS bucket** ? the engine
- [x] A bucket video ? (engine processes) ? the **SPA loads its rallies** ? select ? compile ?
**reel downloaded to the local machine**, all driven from the UI (Path A allows a one-line CLI
pre-process). ? ? **Path A done (local, 2026-06-21)**, see §7.
- [x] The **same flow works with the UI in Hindi** ? multilingual workflow confirmed. ? ?
(Kannada also works via the same switcher.)
- [x] *(Path B = B-P2, SPA-only)* the bucket video is pointed-at **from the UI** (no CLI pre-
step). ? ? **DONE (live e2e, 2026-06-21), see §9.** In the SPA: entered a `gs://` URI ? `POST
/api/process` ? polled `/api/jobs` ? **12 rallies loaded** ? Build ? **reel downloaded (1080p,
76s)**, all from the UI. Surfaced + fixed an engine compile bug along the way (engine
[#289](https://github.com/Khesutra/badminton-highlight-indexer/pull/289)). **Screencast:**
`gs://khesutra-rally-corpus/promo/khesutra-bucket-cuj-screencast.mp4`.
-- [ ] *(option 3)* the same CUJ confirmed on the **CPU droplet** (Linux deploy target) ?
droplet reachable but currently **un-provisioned** (no engine checkout; ~3.8GB RAM) ? needs a
from-scratch engine install or a proper deploy of the fixed engine.
+- [ ] *(option 3, re-scoped 2026-07-02)* the same CUJ confirmed on a **Linux VM** (Linux deploy
target) ? the CPU droplet this originally targeted was **deleted 2026-06-25** (PR #117); re-run
on any future VM stand-up instead.

```

```

## 7. ? Run result ? Path A, local (2026-06-21)
Ran the real CUJ end to end on the local Windows box, isolated engine off `origin/master`
(fresh DB/output worktree), **Gemini** detector:
diff --git a/docs/LOCAL_APPLIANCE_ARCHITECTURE.md b/docs/LOCAL_APPLIANCE_ARCHITECTURE.md
index 22d29d0..e9287ba 100644
--- a/docs/LOCAL_APPLIANCE_ARCHITECTURE.md
+++ b/docs/LOCAL_APPLIANCE_ARCHITECTURE.md
@@ -45,7 +45,7 @@ The decoupled engine just shipped a `StorageBackend` ABC + a `python -m
backend.
- **Compute seam:** `indexing.detector_impl` ? `{`auto`=WSL, `native`=Linux-native, `stub`=CPU
mock}; default `auto` `[verified models.py:129]`, resolved in
`trajectory_hybrid._resolve_runner` `[verified :76-85]`. **Native WASB `native_wasb_runner.py`
is now on `origin/master`** (no longer branch-only; `tests/test_native_wasb_runner.py` is on
master too) `[verified git ls-tree origin/master]`; **`fusion_hybrid._build_native_runner`
returns `NativeWasbRunner`** `[verified fusion_hybrid.py:86, origin/master]` while
**`trajectory_hybrid._build_native_runner` still raises `NotImplementedError`** `[verified
trajectory_hybrid.py:61, origin/master]`. Acceptance gate = **owner byte-parity** (WSL CSV ==
native CSV) on a real GPU box ? not exercisable on the droplet or in the offline suite. Treat
native as **not production-verified**.
- **Worker CLI:** `python -m backend.worker {infer|train}` pulls a `--video-uri` ? pipeline ?
pushes `outputs/<video_id>/{intervals.csv,report.json}` `[verified worker/`__main__`.py:88,149]`.
The validated GPU-free/key-free combo: `detector_impl=stub` + `--set
indexing.skip_ai_handoff=true` + `storage.s3`.
- **Gating:** cloudflared Tunnel (outbound-only, no inbound ports) + Cloudflare Access email-
OTP; backend trusts `cf-Access-Authenticated-User-Email` `[HOSTING_PLAN.md]`. `GET /api/health`
**now EXISTS** (engine M1, [#268]) ? `{status, sport, default_segmenter}` `[verified
main.py:127, corrected 2026-06-21 ? was "does not exist"]`.

```

-- **Droplet reality**: ``168.144.126.176``, **CPU-only**, NO GPU, region ``blr1``; co-hosts the marketing site (``/srv/khelsutra-site``, node ``:8787``) **and** the engine (``~/rally``); ran the full train+infer e2e against DO Spaces with the **GPU mocked** ``[SETUP_NEW_MACHINE.md, DECOUPLED_COMPUTE/HANDOFF.md]``. Shared testbed: MinIO ``9000/9001`` + GCS emulator ``9023`` also run there.

+-- **Droplet reality** (HISTORICAL ? the droplet was decommissioned + deleted 2026-06-25, PR [#117](https://github.com/avidullu/khelsutra/pull/117); data archived to ``gs://khelsutra-rally-corporus/archive/droplet-claude-do-rally-test-20260625/``): **it was** ``168.144.126.176``, **CPU-only**, NO GPU, region ``blr1``; co-hosted the marketing site (``/srv/khelsutra-site``, node ``:8787``) **and** the engine (``~/rally``); ran the full train+infer e2e against DO Spaces with the **GPU mocked** ``[SETUP_NEW_MACHINE.md, DECOUPLED_COMPUTE/HANDOFF.md]``. The shared testbed (MinIO ``9000/9001`` + GCS emulator ``9023``) went away with the box. There is currently **no live engine host** ? the appliance targets "any VM/box", not a specific droplet.

- **Shipped** since this doc's first draft (all on ``origin/master``): ``GET /api/capabilities`` (now also carries a **storage** block ? usable media local/s3/gcs/gdrive + free space, P3) . ``/api/health`` . ``/api/version`` . ``GET /api/videos`` (the "my videos" list, P6) . ``GET /api/source/{id}`` (a frame-identical annotation proxy, D4) . a per-video **cost ledger** (``/api/jobs/{id}/cost``, ``/api/videos/{id}/cost``) . ``POST /api/annotations`` (P8) . ``POST /api/assets`` (store-into-medium, P4) . a **free-space guard** (HTTP 507 on upload/fetch; WSL frame-stage now BLOCKS, P2).

- **Still absent** (net-new): ``GET /api/reels/{video_id}`` (per-video reel list; the global ``/api/reels`` exists but the SPA doesn't consume it), any **Dockerfile** ``[verified ? grep returns none]``.

@@ -173,15 +173,16 @@ Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One small**

Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the start ? owner provisions the host + token; the engine session lands native WASB; the UI MVP does **not** block on it (ships on Gemini).

-**? Alpha deploy path chosen = SPLIT**, not the single-origin shell (owner 2026-06-21). **The alpha ships the split topology** (SPA on Cloudflare Pages + engine on an ``api.`` cloudflared tunnel + CF Access over both ? the engine ``HOSTING_PLAN.md`` S2), **not** the single-origin Caddy shell of **T2** (kept as the documented alternative ? S0/D1). The **deploy kit + runbook** (``[deploy/DEPLOY_ALPHA.md]``(``../deploy/DEPLOY_ALPHA.md``) + ``deploy/cloudflared.config.example.yml``) is shipped (? templates+runbook). **The alpha stand-up is now LIVE** (2026-06-22): **the engine booted on the droplet** (``168.144.126.176``), ``app.khelsutra.guru`` is up, and the full **DNS ? Cloudflare ? Access ? tunnel ? engine** chain plus the **T3** raw-``:8000``-refused posture test (boot-posture fail-fast, Host-guard ``400?200``, path-jail, T4 upload?jail, edge-auth ``401``-gate) were all **exercised live** ``[deploy/droplet/README.md:100-106]``. Only the **Gemini-key rotation** + a **browser OTP login** remain. Enabled by: engine exposure-safety boot posture (engine #294) + SPA ``VITE_API_BASE`` ([#64]).

+**? Alpha deploy path chosen = SPLIT**, not the single-origin shell (owner 2026-06-21) ? since REVERSED: single-origin is now the DEFAULT (see the historical note below). **The alpha shipped the split topology** (SPA on Cloudflare Pages + engine on an ``api.`` cloudflared tunnel + CF Access over both ? the engine ``HOSTING_PLAN.md`` S2), **not** the single-origin Caddy shell of **T2** (kept as the documented alternative ? S0/D1). The **deploy kit + runbook** (``[deploy/DEPLOY_ALPHA.md]``(``../deploy/DEPLOY_ALPHA.md``) + ``deploy/cloudflared.config.example.yml``) is shipped (? templates+runbook). **The alpha stand-up went LIVE** 2026-06-22: **the engine booted on the droplet** (``168.144.126.176``), ``app.khelsutra.guru`` was up, and the full **DNS ? Cloudflare ? Access ? tunnel ? engine** chain plus the **T3** raw-``:8000``-refused posture test (boot-posture fail-fast, Host-guard ``400?200``, path-jail, T4 upload?jail, edge-auth ``401``-gate) were all **exercised live** ``[deploy/droplet/README.md`` ? the 2026-06-22 "Live-verified" note]. Enabled by: engine exposure-safety boot posture (engine #294) + SPA ``VITE_API_BASE`` ([#64]).

+**HISTORICAL** (recorded 2026-07-02): that droplet was decommissioned + deleted 2026-06-25 (PR [#117](https://github.com/avidullu/khelsutra/pull/117); data archived to ``gs://khelsutra-rally-corporus/archive/droplet-claude-do-rally-test-20260625/``), so the alpha is **paused pending the Cloud-Run move** ? nothing is live today. The durable topology decision that came out of the live run: **single-origin is the default for SPA + CF-Access** ? the split ``app.``/``api.`` shape broke the Access session/sign-in flow (issues [#97](https://github.com/avidullu/khelsutra/issues/97), [#114](https://github.com/avidullu/khelsutra/issues/114)); the 2026-06-22 fix repointed serving to a single origin (see ``deploy/DEPLOY_ALPHA.md``'s status banner). A future stand-up should

start from the single-origin shell (\$0/D1), not this split kit.

```
## 10. Open questions ? owner / external
**Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs
cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention
defaults** ? **window DECIDED** (uploads **7d**, reels durable; [engine
#299](https://github.com/Khelsutra/badminton-highlight-indexer/pull/299) tool + scheduling
[#301](https://github.com/Khelsutra/badminton-highlight-indexer/issues/301)); **still open:**
auto-delete originals after compile vs keep proxies+intervals only (privacy/consent). (3) **per-
video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4) **GPU host choice** ? DO GPU
droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker worth building for Cloud
Run)? (5) **P5 bridge ownership** ? who builds the orchestration endpoint + how it coordinates
with the active worker sessions.
**Nice-to-knows:** console home (web/ vs engine static); a review-playback streaming endpoint;
per-user scoping timing; containerization vs runbook.
```

```
## 11. Definition of done
-- [~] CUJ-1 works end to end on the droplet (Gemini **and** stub) behind the gate, via the
ingest menu. *(2026-06-22: chain LIVE end-to-end ? DNS?Access?tunnel?engine, T4 upload?jail
exercised; the full upload?reel pass still needs the Gemini key + a browser OTP login
`[deploy/droplet/README.md:100-106]`.)*
-- [x] Raw `:8000`/`:9000` unreachable from outside (**tested** live 2026-06-22, not assumed)
`[deploy/droplet/README.md:100-106]`.
+- [~] CUJ-1 works end to end behind the gate, via the ingest menu (Gemini **and** stub).
*(2026-06-22: chain proven LIVE end-to-end on the droplet ? DNS?Access?tunnel?engine, T4
upload?jail exercised; the full upload?reel pass never got its Gemini key + browser OTP login
before the droplet was deleted 2026-06-25 `[deploy/droplet/README.md ? the 2026-06-22 "Live-
verified" note]`. Needs a re-run on the next stand-up.)*
+- [x] Raw `:8000`/`:9000` unreachable from outside (**tested** live 2026-06-22 on the droplet,
not assumed) `[deploy/droplet/README.md ? the 2026-06-22 "Live-verified" note]`.
- [ ] Local upload wired; bucket-URI ingest at least designed/tracked; Drive scoped honestly
(not faked).
- [ ] RallyPicker embedded with honesty rules; `min_shot_count` surfaced.
- [ ] `/api/capabilities` + `/api/health` + banner truthful; reload-survival via `/api/videos`
+ `/api/reels`.
@@ -194,6 +195,6 @@ Legend: ? Todo . ? In progress . ? Done . ? Blocked/gated. **One smal
```

```
---
### Changelog
+- 2026-07-02 ? Droplet-retirement pass: S3 "Droplet reality" and the S9 alpha stand-up note
marked **HISTORICAL** (droplet decommissioned + deleted 2026-06-25, PR #117; data archived to
`gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/`); recorded the
durable **single-origin-default** decision for SPA + CF-Access (#97/#114);
`deploy/droplet/README.md` citations re-anchored (line numbers shifted by its retirement
banner).
- 2026-06-22 ? Freshness pass: `DRAFT`?`IN PROGRESS`; re-anchored engine facts against
`origin/master` ? `/api/capabilities` + `/api/health` now exist (T7 engine half ?);
`native_wasb_runner.py` is on master (no longer branch-only); CORS `` is now env-lockable + the
engine ships a Host-guard / optional Cf-Access edge-auth / boot-posture (so "zero app-layer
auth" is no longer literal); alpha deploy went LIVE 2026-06-22 (`app.khelsutra.guru`), ticking
the raw-`:8000`-refused DoD box.
- 2026-06-20 ? Initial `DRAFT`. Consolidated three single-box-appliance designs (FastAPI-native
/ unified-origin / decoupled-profile) into the reverse-proxied unified-origin spine + compute-
as-a-setting framing, via a multi-agent design pass; every engine-fact anchor re-verified
against source before writing (catching the native-WASB partial-wiring + `ai_max_workers`
default nuances). Folded in the owner's ingest-menu (Drive/bucket/local), multi-provider GPU,
and first-class Observability directives.
```

56. assistant (2026-07-02T04:57:50.441Z)

The diff is complete and correct. Now commit ? re-verifying branch state first, staging explicit paths only.

57. user (2026-07-02T04:58:05.393Z)

```
docs/retire-droplet-doc-rot
f9eea14 web: resilient video-summary readers + show fps in the library (#140)
[docs/retire-droplet-doc-rot bf11676] docs: record the droplet retirement ? DEPLOY_ALPHA
retired, stale LIVE claims swept (R1-06/09/35/36)
 6 files changed, 117 insertions(+), 64 deletions(-)
bf11676 docs: record the droplet retirement ? DEPLOY_ALPHA retired, stale LIVE claims swept
(R1-06/09/35/36)
```

58. user (2026-07-02T04:58:15.146Z)

```
remote:
remote: Create a pull request for 'docs/retire-droplet-doc-rot' on GitHub by visiting:
remote:      https://github.com/avidullu/khelsutra/pull/new/docs/retire-droplet-doc-rot
remote:
branch 'docs/retire-droplet-doc-rot' set up to track 'origin/docs/retire-droplet-doc-rot'.
To https://github.com/avidullu/khelsutra.git
 * [new branch]      docs/retire-droplet-doc-rot -> docs/retire-droplet-doc-rot
```

59. user (2026-07-02T04:58:43.413Z)

<https://github.com/avidullu/khelsutra/pull/141>

60. assistant (2026-07-02T04:58:44.091Z)

You've hit your session limit · resets 1:10pm (Asia/Calcutta)